INTEGRATED PRACTICE QUIZ:

Pharmacology of Aminoglycosides, Protein Synthesis Inhibitors, Folate Synthesis Inhibitors, and Fluoroquinolones

This quiz has lots of questions because it covers four major antibiotic classes and various frequently occurring infections in various real-life contexts.

HINT: Identify the clues in the vignette that you need to answer the question. Don't be distracted by the information that is not needed to answer the question.

- **Suggestion for solving the puzzles:** Circle the clues and think about each option – reason how each option fits the clues and how it doesn't fit the clues.

  It is just as important to understand why the correct option(s) is correct and the wrong options are incorrect.

  Then, when the answers are posted, look at the explanations. At a later time (spacing practice), try it again without looking at the explanations.

Tips for successful learning:

The *purpose* of integrating four classes of antibiotics into one practice quiz is to *help you improve your knowledge and understanding of new material* by mixing up problem types ("interleaving"), rather than focusing on one type at a time before moving on to study another type.

Self-quizzing, spaced retrieval practice, interleaving, and other study practices that solidly embed knowledge help learners avoid the "illusion of knowing".

When will the answers be posted?

Practice points included on the answer key in some of the rationales are intended to give you context, provide relevance, and help your deeper understanding of drug usage.

The answers will be posted on the Friday before Thanksgiving break.

Meanwhile, all answers can be found by doing a search of the relevant Notes handouts.

Abbreviations:

Sig = signetur translated as "let it be labeled"

po = per os (by mouth); IM = intramuscular; IV = intravenously; IT = intrathecally; IO = intraosseously

Q12h= every 12 hours | Q8h= every 8 hours | TID= three times daily | BID= twice daily | QD= once daily

Learning Tip: Retrieval Practice means self-quizzing (before looking at the answers)
Self-quizzing by focusing on the main ideas and the meanings helps you to focus on the central principles and provides ***a reliable measure*** of what you have learned and what you have not yet mastered.

The quiz begins on the next page.

Leslie Goldstein, Pharm.D.
Updated September 2025

1. Which one of the following is a common feature among all the protein synthesis inhibitors?

A. Action at the bacterial ribosome*

B. Bactericidal activity

C. Low toxicity to host cells

D. PK-PD profiles

E. Renal elimination

F. Spectrums of activity

Rationale: A. action at the bacterial ribosome

The protein synthesis inhibitors bind to the bacterial ribosome and interfere with protein synthesis. The sites of action vary among the different classes.

Incorrect options:

- B. Activity: Except for the aminoglycosides, the protein synthesis inhibitors have mainly bacteriostatic activity. (Recall that aminoglycosides are bactericidal.)
- C. Toxicity: Unlike the beta-lactam antibiotics, which have low toxicity to host cells, each of protein synthesis inhibitors classes are associated with their own characteristic adverse effects. Most are manageable, but caution, monitoring, and informing patients are required.
- D. PK-PD profiles: The dosing schedules that maximize antibiotic effect (PK-PD, profile) vary among the drugs and the susceptible microorganisms.
- E. Elimination: Modes of clearance vary.
- F. Spectrums of activity: The microorganisms' susceptibility to the different antibiotics varies.

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2. What is the mechanism by which aminoglycosides gain access to bacterial cell cytoplasm?

A. Active transport by oxygen-dependent process*

B. Active transport by iron-transport systems

C. Endocytosis

D. Passive diffusion through cytoplasmic lipid bilayer

E. Passive diffusion via a solute carrier

Rationale:

Aminoglycosides are actively transported across the bacterial cell membrane by an oxygen-dependent process, which must be functioning for these drugs to be effective. Therefore, aminoglycosides are ineffective against anaerobic bacteria.

What drugs enter cells by the mechanisms in the other options?

- Active transport by iron-transport systems: Cefiderocol is a "siderophore" (iron chelator). Cefiderocol chelates ferric iron, allowing the drug to cross the outer membrane of Gram-negative bacteria using iron transport systems.
- Endocytosis: Antibiotics do not enter bacterial cells by endocytosis, which is a mechanism of eukaryotic cells to engulf substances.
- Passive diffusion through cytoplasmic lipid bilayer: Some antibiotics are lipophilic and can passively diffuse across bacterial membranes (and possibly host cell membranes) by passive diffusion.

Examples from this series of lectures are chloramphenicol, ciprofloxacin, levofloxacin and moxifloxacin.

– Passive diffusion via a solute carrier: Solute carrier proteins (SLCs) are eukaryotic transporters. Bacterial do express SLC-like transporters.

  Looking ahead: We may see antibiotics designed in the future that take advantage of bacterial transporters to overcome permeability barriers, especially in Gram-negative bacteria.

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3. Why are aminoglycosides effective in the treatment of susceptible infection with extended-interval once daily dosing even though the elimination half-life in adults with normal renal function is only about 2-3 hours?

Explain the actions and effects using the PK-PD model.

Rationale: PK-PD profile ***predicts efficacy*** of an antibiotic and ***guides dosing*** recommendations for the treatment of infections by specific microbes.

Aminoglycosides exhibit Peak/MIC concentration-dependent killing with significant post-antibiotic effect (prolonged persistent effects).

Aminoglycosides extended-dosing demonstrates efficacy comparable with traditional intermittent dosing and the possibility of decreased nephrotoxicity.

Review: PK-PD Profiles – Models of Antimicrobial Efficacy

Concentration-dependent bactericidal effect:

- Peak/MIC (C_{max}/MIC): Higher peak concentration leads to faster, more extensive rate and extent of bacterial killing.
  - Significant post-antibiotic effect (PAE) – suppression of bacterial growth after drug clearance – allows extended-interval dosing.
  - Aminoglycosides.
- AUC/MIC: Total cumulative exposure of the bacteria to the antibiotic over time best correlates with favorable clinical outcomes and minimizing the development of resistance.
  - Vancomycin, fluoroquinolones, doxycycline/minocycline, macrolides, daptomycin, tedizolid

Time-dependent bactericidal effect: The longer the drug concentration stays above the MIC during the dosing interval, the more effective the antibiotic is at killing the bacteria.

- These antibiotics do not produce a greater rate and extent of bacterial killing (better efficacy) with high peak concentrations and do not have clinically important post-antibiotic effect.
  - Beta-lactams, linezolid

NOTE: Some antibiotics exhibit a mixed PK-PD profile, meaning their efficacy is best predicted by more than one parameter.

For example, for most antibiotics, AUC/MIC ratio is the best predictor of efficacy against specific pathogens. Certain drugs may also exhibit a greater rate and extent of bacterial killing with higher peak/MIC ratio. For others, efficacy is also predicted by concentrations above the MIC for extended durations (T>MIC).

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4. Which organisms would be intrinsically resistant to the aminoglycosides and by what mechanisms?

Short answer

Rationale:

Obligate anaerobic organisms
Organisms in an anaerobic environment
Gram-positive bacteria with an intact cell wall

AMINOGLYCOSIDE: WHAT YOU NEED TO KNOW

PK	Hydrophilic, IV/IM/IT, glomerular filtration and renal excretion of unchanged drug
PK-PD profile	Concentration-dependent: Peak/MIC
Mechanism of action	Enter bacteria via porins and oxygen-dependent transport across cell membrane Bind 16S rRNA in 30S ribosome: 1)interfere with initiation complex, 2) induce misreading and incorporation of incorrect amino acids, 3) block translocation on mRNA, 4) induce premature termination, 5) possibly: breaking up of polysomes to monosomes and aberrant proteins may cause membrane damage
Therapeutic Uses	Serious aerobic Gram-negative infections (sepsis, pneumonia, urinary tract infections, intra-absominal infections) Infective endocarditis, streptococcal or enterococcal, combined with a cell wall inhibitor (typically ampicillin or vancomycin) for synergy

AMINOGLYCOSIDES RESISTANCE MECHANISMS	
Intrinsic Resistance Mechanisms	
Strict (obligate) anaerobic organisms	They lack the oxygen-dependent active transport mechanism necessary for AGs to be actively transported across the bacterial cell plasma membrane. Therefore, AGs do not accumulate inside the bacterial cell.
Facultative anaerobes in an anaerobic environment	
Gram-positive bacteria:	Aminoglycosides may not able to traverse the intact thick peptidoglycan cell wall. AGs have effective activity against Enterococcus, Streptococcus and Staphylococcus only in combination with a cell wall synthesis inhibitor for synergy in the treatment of infective endocarditis.
Acquired resistance mechanisms	
Drug inactivation The principle mechanism of resistance	Synthesis of transferase enzymes that modify and inactivate aminoglycoside antibiotics by ***phosphorylation***, ***adenylation***, or ***acetylation*** of a specific hydroxyl or amino group on the drug molecule.
Over 50 different transferases have been identified. The genes encoding the aminoglycoside modifying enzymes are usually found on plasmids and transposons. (Jawetz, et.al., 27e; Chapter 14,15)	

Impaired entry	– Alteration or deletion of porin proteins involved in aminoglycoside transport – Nonfunctional oxygen-dependent transport process, as in anaerobic conditions – Efflux pumps. *Pseudomonas aeruginosa*
Binding site modification	Point mutations in the 16S rRNA that change the structure of the binding sites and reduces affinity for the drugs. – Examples: *Mycobacterium tuberculosis* resistance to streptomycin and *E. coli* resistance to gentamicin.
	Ribosomal methylation of the 16S rRNA (plasmid transmission) – Examples: Various Gram-negative species, including *Pseudomonas, E. coli, Klebsiella pneumoniae, Proteus mirabilus, Acinetobacter baumannii*
Resistance patterns vary by bacterial species and local epidemiology. Most strains of Enterbacterales have acquired resistance to streptomycin. ***Enterococcus faecalis* and *E. Faecium*:** The enterococcal aminoglycoside-modifying enzyme (bifunctional phosphotransferase / acetyltransferase) inactivates gentamicin tobramycin, amikacin and netilmicin (cross-resistance), *but not streptomycin*, which has differences in its chemical structure **Cross-resistance:** Gentamicin resistance predicts resistance to tobramycin. Amikacin and Plazomicin may retain activity.	

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5. A 10-year-old boy presents to the pediatrician with complaints of headache, fatigue, malaise, and muscle and joint aches for the past 5 days, and "a big bug bite" on his right leg. He and his friend were exploring in the woods outside of their Long Island town the weekend before last (10 days ago). NKDA.

UpToDate: Erythema migrans Graphic 81270 Version 1.0 accessed from Lyme disease graphics.

Fever is absent and vital signs are normal. A single circular red area that clears in the middle, forming a bullseye pattern (erythema migrans), at the site of the tick bite is noted on physical exam. Lymphadenopathy is noted in the regional lymph nodes.

The patient is given a clinical diagnosis of Lyme disease. The patient states that he can swallow tablets. An e-prescription is submitted for a ten-day course of antibiotic therapy with an oral broad-spectrum drug effective in the treatment of many zoonotic diseases.

The patient and his mother are advised that the drug can cause stomach upset and can be taken with food a liquid and the patient should not lie down for at least 30 minutes after taking the medicine to prevent irritation of "your food pipe, which is call the esophagus". They are also cautioned about wearing long sleeves, long pants, a hat, and sunblock while in the sun as the drug may cause a bad sunburn-like reaction (photosensitivity).

i) Which one of the following options best represents the mechanism of action of the selected antibiotic?

A. Prevents access of aminoacyl tRNA to the acceptor (A) site on the mRNA-ribosome complex*
B. Prevents assembly of the functional ribosome and causes misreading of the genetic code

C. Prevents formation of the ribosomal-fMet-tRNA complex that initiates protein synthesis
D. Prevents the interaction between peptidyl transferase and its amino-acyl tRNA
E. Prevents the translocation of a newly synthesized peptidyl tRNA molecule from the acceptor site (A) to the peptidyl (P) donor site

Hint: Think about the cause of the infection and the group of drugs that are very effective against this classification of diseases. (Hint: key word: zoonotic)

ii) Where is the antibacterial binding site of this drug?

iii) What is the organism that causes Lyme disease?

iv) What treatment would be recommended if the child is 6 years old?

Rationale: A. A-site

Doxycycline is the most frequently used drug of the tetracyclines subclass of protein synthesis inhibitors. Tetracyclines (TCNs) enter microorganisms by 2 modes: passive diffusion, and an energy-dependent mechanism.

i) **Mechanism of action:** TCNs bind the 16S rRNA on the 30S ribosome and block the amino acid-charged tRNA (amino acyl-tRNA) from binding to the acceptor A site on the mRNA complex, which prevents addition of the amino acid to the growing peptide chain, thus halting protein synthesis.

ii) **Drug-food interaction:** Di- and trivalent cations chelate the drugs in the tetracyclines group, reducing absorption. Food impairs absorption of tetracycline, demeclocycline, and omadacycline so these drugs should be taken on an empty stomach. Doxycycline has a low affinity for calcium and may be taken with food, including dairy products.

iii) **Microbiology:** Lyme disease is caused in the US by the spirochete *Borrelia burgdorferi*, which is transmitted by the bite of the *Ixodes* tick (deer tick or black-legged tick) (pronunciation: ick-so-dese) *I. scapularis.*

Lyme disease is common in the Northeast, mid-Atlantic, and upper-Midwest regions.

iv) **Clinical applications:** The American Academy of Pediatrics permits the use of doxycycline for ≤21 days in children of all ages.

➢ **Practice points:**
- Goal of therapy of early Lyme disease:

  To shorten the duration of signs and symptoms and prevent progression to later stages of Lyme disease.
- Doxycycline *is the preferred oral tetracycline for most indications*:

  It is taken twice daily,
  is generally well tolerated, and
  absorption is not significantly affected by food.

  Doxycycline is excreted in the feces. Therefore, renal dosing is not required for patients with renal impairment.

GLYCYCYCLINES in brief:

Tigcycline is starred(✳), the first and most widely used.

Derived from tetracyclines

Broad spectrum

Designed to overcome resistance mechanisms

Spectrum: Gram-positive and Gram-negative

Complicated skin/skin structure infections, complicated intra-abdominal infections, community-acquired pneumonia (*S. pneumoniae, H. influenzae, Legionella pneumophila*)

What drugs are represented by the mechanisms in the other options?

Incorrect options:

- B. **Aminoglycosides** interfere with the initiation complex of peptide formation, cause misreading of the mRNA with incorporation of incorrect amino acids into the peptide chain, and breakup the polysomes into nonfunctional monosomes.
- C. **Oxazolidinones**, linezolid and tedizolid, have a unique binding site on the 23S rRNA of the 50S ribosome, which prevents the formation of the 70S ribosome, which prevents initiation of protein synthesis.
- D. **Chloramphenicol'**s binding site is on the 50S ribosome and it inhibits peptide bond formation.
- E. **Macrolides** bind to the 23S rRNA of 50S ribosome near the center of the peptidyl transferase, which blocks the enzyme's polypeptide exit tunnel inhibits, which inhibits the translocation of the new peptidyl-tRNA to the peptidyl site, resulting in the release of incomplete proteins.

<table>
<tr><td colspan="2">DOXYCYCLINE: WHAT YOU NEED TO KNOW
Other drugs in the class: Tetracycline, minocycline, demeclocycline, glycylcyclines and advanced generation TCNs
Takeaways: Most commonly used TCN, broad spectrum, zoonotic infections, twice daily dosing, nonrenal elimination</td></tr>
<tr><td>PK
· Oral, IV; not metabolized; biliary / fecal excretion unchanged drug
PK-PD Profile: Mixed
Concentration-dependent AUC/MIC with moderate PAE (up to 3 hours)
Time-dependent, T>MIC: The drug also must be above the MIC for a certain amount of time for maximal effectiveness
Mechanism of action
· Enters bacteria by passive diffuse through porins, then active transport across the inner membrane
· Binds 30S subunit → blocks access of amino acyl-tRNA to A site → prevents addition of amino acids to growing peptide chain
Spectrum: Bacteriostatic
· Broad spectrum
· BUT widespread resistance to Gram-positive bacteria and many Gram-negative strains</td><td>Resistance mechanisms
· G+: Mainly by Tet(K) efflux pump and Tet(M) ribosomal protection proteins
· G-: Mainly by Tet(AE) efflux pump.
Therapeutic uses (currently)
· Zoonotic infections, atypical bacteria, and susceptible MRSA – effective and preferred
· Short-term (≤21 days) for treatment children <8 years and pregnant people is now recommended.
TOXICITY
· Nausea, vomiting, photosensitivity, ↑blood urea nitrogen (BUN), hepatotoxicity (dose-dependent, rare, may be fatal)
BUN is a marker of kidney function. Urea is a biproduct of protein metabolism that is filtered by the glomerulus and excreted in urine.</td></tr>
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6. Patients are cautioned to avoid di- and trivalent cation-containing foods and supplements when taking an oral drug from the tetracycline class.

Why?

Rationale:

i) Di- and trivalent cations chelate some TCNs in the gut forming poorly soluble complexes. The absorption of tetracycline, omadacycline, and demeclocycline is reduced.

Doxycycline and minocycline bind calcium less avidly than other TCNs. They are partially inactivated by chelation.

Concomitant antacids and nutritional supplements containing di-, trivalent cations should be avoided.

Advise patients, that if antacids or mineral supplements are needed, take them at least 2 hours before or after taking the antibiotic. This caution applies to all oral drugs in the tetracyclines class.

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7. KC Jr. is 38-day-old uncircumcised male infant brought to the pediatrician due to fever of 10 hours, irritability, and poor feeding. He was delivered vaginally at 36 weeks gestational age (premature) and has been thriving since being discharge home with his parents. Temperature (rectal) is 100.8°F. All other physical findings are within normal limits. NKDA. A dipstick urinalysis is positive for leukocyte esterase and nitrite consistent with an *E. coli* infection. A urine sample is sent to the microbiology laboratory for culture and sensitivity testing. A sulfonamide antibiotic would likely be effective but should be avoided in this patient who is less than 2 months old.

What is the potential adverse effect that could occur with sulfonamide use in this patient?

A. Organ damage due to accumulation of toxic drug metabolites
B. Tooth discoloration due to chelation with calcium in tooth enamel
C. Elevated plasma levels of free unconjugated bilirubin due to displacement from albumin binding sites*
D. Hemolytic anemia due to glucose-6-phosphate dehydrogenase deficiency
E. Renal tubule irritation due to crystalluria

Rationale: C. elevated plasma bilirubin

Mechanism of adverse drug event: Elevated plasma free bilirubin levels.

Because of immature hepatic metabolism capacity, especially UGT, and the immature blood-brain barrier, the free unconjugated bilirubin in circulation can deposit in brain tissue.

Physiology: Bilirubin

Bilirubin is produced in the breakdown of heme in hemoglobin.

Heme is converted to biliverdin, then reduced to unconjugated bilirubin (lipophilic), which binds albumin in the bloodstream for transport to the liver.

Unconjugated bilirubin is conjugated with glucuronic acid by UDP-glucuronosyltransferase (UGT1A1).

The water-soluble conjugated bilirubin is secreted into the bile.

Gut bacteria in the large intestine convert it into urobilinogen, which is excreted in the urine as urobilinogen and the feces as stercobilin.

Know this

- **Practice point:** Sulfonamide therapy is not recommended in infants ≤2 months old (48 weeks gestational age).

In the young infant, free unconjugated bilirubin can pass into the CNS and cause kernicterus

SULFONAMIDES: WHAT YOU NEED TO KNOW	
Trimethoprim-sulfamethoxazole (TMP-SMX) is the only one currently used in the US for treatment of systemic bacterial infections. Takeaways: PK, MOA, sulfamethoxazole in fixed combination with trimethoprim for synergy. Widespread resistance is seen in many Gram-positive and Gram-negative species.	
PK: Distributes to fluids and secretions; hepatic metabolism; excreted in urine and feces **PK-PD profile:** AUC/MIC (limited data) **Mechanism** · Sulfamethoxazole: Competitive inhibitor of dihydropteroate synthase · Trimethoprim inhibits bacterial DHFR · Prevent folate synthesis, thus, DNA, RNA, and protein synthesis **Spectrum of activity** · Aerobic Gram-positive organisms, including MRSA · Some Gram-negative aerobes, including *E. coli, Klebsiella, Proteus, Haemophilus influenzae* · ***Not effective*** against atypicals or *Pseudomonas* · Resistance is common. **Therapeutic uses** · **First-line** for Pneumocystis pneumonia – an AIDS-defining fungal infection that is an indicator of advance HIV. An option for uncomplicated urinary tract, skin/skin structure, and upper respiratory infections caused by susceptible organisms	**TOXICITY** · **Infants:** Displaces bilirubin on albumin binding sites – may cause kernicterus in young infants and potential for drug interactions · **G6PD deficiency:** Dose-related potential for hemolytic anemia in patients with G6PD deficiency – Severity of hemolysis varies with degree of severity of the deficiency and the particular sulfonamide. Individual risk may depend on the specific G6PD variant and clinical context. · **Hypersensitivity reactions** – skin rash and systemic symptoms – fever, SJS/TEN, hepatoxicity, blood dyscrasia · **Crystalluria nephrotoxicity** (acidic pH) **Rare serious toxicities** · Aplastic anemia, rare, irreversible, life-threatening · Folate deficiency

(encephalopathy) with neurological dysfunction ranging from mild to severe and irreversible. Premature neonates are at greatest risk but it can occur in term infants.

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8. KC Jr. is evaluated for outpatient treatment. *E. coli* resistance to ampicillin is prevalent in the region. An e-prescription is submitted for the oral suspension of a narrow-spectrum beta-lactam from a different subclass. The drug has good bioavailability and good activity against the infective pathogen: Sig: 75 mg/kg po TID x10 days.

What is the drug?

Big hints: 1) narrow spectrum, 2) the mnemonic for the drug's Gram-negative activity is PEcK.

FYI: You do not need to know dosing for now. You might find it helpful to get a sense of how drugs are prescribed.

(UpToDate Algorithm: Choice of empiric antibiotic to treat urinary tract infection in children 1 month to 2 years.)

Rationale: Cephalexin

Cefazolin (parenteral) and cephalexin (oral) are first-generation cephalosporins with a narrow spectrum of action but good activity against aerobic streptococci and staphylococci (MSSA), and the Enterobacterales that are frequent causes of urinary tract infections: Proteus mirabilis, E. coli, and Klebsiella pneumoniae – PecK.

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9. A 17-year-old female presents to the dermatology clinic with moderate inflammatory acne affecting her face and upper back. She has tried over-the-counter benzoyl peroxide and topical retinoids with limited success. The dermatologist prescribes an oral antibiotic in the tetracyclines class as part of her treatment plan.

After six weeks of therapy, her acne has improved significantly. However, she reports experiencing occasional dizziness and a dark discoloration on her gums.

What is the name of the drug?

A. Demeclocycline
B. Doxycycline
C. Minocycline*
D. Tetracycline
E. Tigecycline

Rationale: C. Minocycline

Minocycline highly lipophilic so it penetrates into sebaceous glands and skin tissue.

It is effective against *Cutibacterium acnes* (formerly *Propionibacterium acnes*) that is an important contributor to acne Inflammation.

Minocycline has anti-inflammatory properties and helps decrease redness, swelling, and lesion formation.

It convenient, tolerable, and reasonably priced.

Side effects include:

– GI upset: Can take with food
– Dizziness / vertigo: CNS penetration
– Skin pigmentation: blue-gray discoloration of skin, gums, sclera
– Photosensitivity: Sunburn risk
– Drug-induced lupus: rare autoimmune reaction

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10. A 16-year-old male presents to the ED with sudden onset of fever, chills, anorexia, and malaise. He also had a headache, chest soreness, abdominal pain and vomiting earlier today but these symptoms have abated. He went rabbit hunting 3 days ago with his dad. The patient was in charge of cleaning and preparing the rabbits for cooking.

The patient’s temperature is 104°F; a chancre-like ulcer with raised margins is noted on the dorsum of the fourth digit of his right hand and regional axillary lymphadenopathy is noted (see photo).

Blood samples are drawn for serology for suspicion of *Francisella tularensis* (a non-enteric aerobic Gram-negative coccobacillus).

The patient is told that “rabbit fever” can be serious and the treatment associated with excellent rates of cure and minimal relapses by shots into the muscle of the buttocks or thigh every 12 hours for 10 days. A nurse will come to the home twice daily to administer the intramuscular injections, take blood samples for monitoring kidney function and drug levels, and monitor side effects, which include injection site reactions and nerve toxicities. The patient is advised to report ringing in his ears (tinnitus), dizziness (vertigo), tingling or numbness in his extremities, and diminished vision or a blind spot (scotoma).

Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. Mcmichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition

Tularemia. A chancre-like ulcer with raised margins on the dorsum of the fourth digit with accompanying axillary lymphadenopathy. The rash on the chest is unrelated and is pityriasis versicolor.

Citation: Chapter 156 Miscellaneous Bacterial Infections with Cutaneous Manifestations, Kang S, Amagai M, Bruckner AL, Enk AH, Margolis DJ, McMichael AJ, Orringer JS. Fitzpatrick's Dermatology, 9e; 2019. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2570&sectionid=210431229 Accessed: November 06, 2020

Which of the following classes best fits the description of the administered drug?

A. Aminoglycosides*
B. Beta-lactams
C. Chlorampenicol
D. Cyclic lipopeptides
E. Folate synthesis inhibitors
F. Fluoroquinolones
G. Glycopeptides
H. Lincosamides
I. Oxazolidones
J. Macrolides
K. Tetracyclines

Rationale: A. Aminoglycosides

The described drug is streptomycin, which has traditionally been the preferred aminoglycoside for moderate and severe tularemia because of its high efficacy, extensive experience with use, and FDA approval for tularemia.

It’s mechanism of action and side effects are typical of the class.

Streptomycin-specific adverse effects: Neurotoxic effects, paresthesia and scotomas; Injection site can become hot and tender.

Streptomycin is rarely used, not readily available, and very expensive.

- **Practice point:** Gentamicin IM or IV is also recommended for moderate and severe tularemia because:

It is equally effective,
More readily available and inexpensive,
Timely blood levels are more readily available, and
Gentamicin has less ototoxicity.

Gentamicin IM or IV 5 mg/kg/day in 1 to 3 divided doses IM or IV q8h x7-10 days

What are the recommendations for mild to moderated *F. tularensis* infection?

Oral therapy:

Other alternatives for tularemia treatment, outpatient or inpatient:

Ciprofloxacin or levofloxacin or doxycycline.

Post-exposure prophylaxis (people at high-risk for infection): oral ciprofloxacin or levofloxacin or oral doxycycline

- ❖ **Practical considerations: It's good to be aware of alternatives.**
  Some patients may not be able to tolerate the preferred drug or if a drug is unavailable (there are numerous drug shortages).

Why is it important to know about tularemia?

(*Francisella tularensis*, aerobic, fastidious Gram-negative coccobacillus)

- ➢ Tularemia is a disease that can infect animals and people.
- – Untreated *pulmonary* and *typhoidal* tularemia have mortality rates of approximately 35%.
- – Without antibiotics, ulceroglandular disease may last weeks and has a mortality rate of approximately 5%.
- – Uncomplicated recovery is expected with antibiotic therapy. (Fitzpatrick's)
- – Rabbits, hares, and rodents are especially susceptible and often die in large numbers during outbreaks. People can become infected in several ways, including:
  - Tick and deer fly bites
  - Skin contact with infected animals
  - Drinking contaminated water
  - Inhaling contaminated aerosols or agricultural and landscaping dust

Resources: Access Medicine Fitzpatrick's Dermatology, 9e, 2019; Chapter 156: Miscellaneous Bacterial Infections with Cutaneous Manifestations; Figure 156-2 Tularemia

UpToDate: Table Graphic 77460 v. 17.0, Antibiotics for Tularemia

The Sanford Guide to Antimicrobial Therapy, 55e, 2025; Table 1(42) Lung > Tularemia and Table 1(66) Systemic Syndromes > Tularemia

https://www.cdc.gov/tularemia/index.html Page last reviewed Dec 13, 2018

11. A normally healthy 34-year-old woman at 14 weeks gestation presents to the emergency department in Jonesboro, AK with fever, headache, acute abdominal pain, nausea, vomiting, cough, myalgia, malaise, and conjunctivitis. The symptoms came on suddenly last night, about 18 hours ago. She and her husband returned from the Ozark Folk Festival 8 days ago; they stayed at a campground near the event. She does not recall noticing a tick bite.

Blood samples are obtained for indirect immunofluorescence assay testing for *Rickettsia rickettsii*.

Empiric IV antibiotic therapy is initiated based on clinical suspicion of Rocky Mountain spotted fever. The selected drug is metabolized in the liver by glucuronidation and excreted in urine as glucuronide metabolites.

i) What antibiotic was administered to this patient?

ii) Because of potentially serious drug-specific toxicities of the conventional drug, monitoring includes one of the following tests at baseline levels and every 2 days.

The test is:

A. AST and ALT hepatic enzymes
B. Glomerular filtration rate and creatinine clearance
C. Complete blood count*
D. Creatinine kinase and troponin
E. Vitamin B12 levels

iii) What is the first-line drug treatment of this zoonosis?

Rationale: C. complete blood count

i) Chloramphenicol

ii) Serious and fatal blood dyscrasias (aplastic anemia, hypoplastic anemia, thrombocytopenia, and granulocytopenia) have occurred after both short-term and prolonged therapy with chloramphenicol. (Lexidrug monograph: chloramphenicol)

Complete blood count can detect early changes in red cells, white cells, and platelets, allowing early intervention.

iii) Evidence supports doxycycline as the preferred treatment of Rocky Mountain spotted fever. It has high efficacy and significantly reduced mortality when started within the first five days of the illness.

- **Practice Update:** *Doxycycline is now the recommended agent for the treatment of Rocky Mountain spotted fever in pregnant patients and children of all ages.*

  Given the efficacy of doxycycline, the lack of other good alternatives, and the extremely rare incidence of the adverse effects in mother and fetus with doxycycline use, the benefits generally outweigh the risks.

A question related to RMSF in the pregnant patient with **chloramphenicol** as the correct answer could come up on a board exam.

Rocky Mountain Spotted Fever (RMSF) Treatment https://www.cdc.gov/rmsf/healthcare-providers/treatment.html

Incorrect options:

- A. Serum aminotransferases, ALT and AST: Tetracyclines with prolonged therapy
- B. Renal function: Drugs renally excreted in active form require dosage adjustment in patients with impaired renal function.

  Protein synthesis inhibitors that do not require dosage adjustment for reduced renal function are doxycycline, azithromycin, clindamycin, quinupristin-dalfopristin. (Dear Aunt Catherine's Quilt)
- D. Creatine kinase and troponin: Drugs that cause skeletal or cardiac muscle damage may increase serum levels of specific CKs (CK-MM and CK-MB, respectively).

  Daptomycin and statins are two types of drugs that cause skeletal muscle toxicity and elevated plasma CK levels.

  Serum troponin levels are increased in patients who suffer a myocardial infarction.
- E. None of the drugs in the group impact vitamin B12 levels.

CHLORAMPHENICOL: WHAT YOU NEED TO KNOW Takeaways: PK, MOA, Rocky Mountain spotted fever, meningococcal meningitis, toxicities	
PK · IV (succinate salt for solubility in IV fluid), · passive diffusion (lipophilic) and crosses BBB in high concentrations, · hydrolysis to chloramphenicol and UGT metabolism, · excretion of inactive glucuronide metabolites in urine (adults) but 80% as succinate in children · t ½ variable 28-35 hours in premature neonates; ~4 hours in adults **PK-PD profile:** no data **Mechanism of action** · Binds 50S subunit → inhibits peptidyl transferase activity, thereby preventing peptide bond formation **Spectrum:** Broad spectrum · Gram-positive and negative cocci and bacilli; aerobes and anaerobes; atypicals · (*Pseudomonas* is intrinsically resistant.) · Resistance by acetyltransferase inactivation of drug	**Therapeutic uses** · Life-threatening infections only when other agents are not tolerated or available: · Rocky Mountain spotted fever, meningococcal meningitis, typhoid fever **TOXICITY** · Suppression of red cell production, dose-dependent, reversible · Aplastic anemia, rare, irreversible, life-threatening · Gray syndrome (cyanosis), especially neonates (high concentrations) – Chloramphenicol in high concentrations (as in neonates with low UGT function) blocks mitochondrial electron transport, impairing cellular respiration in the liver, heart, and muscle → hypotension, cyanosis, cardiovascular collapse · GI, optic atrophy/neuritis, and CNS – confusion, delirium, depression, headache

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12. A 12-year-old girl presents to the pediatrician with paroxysmal cough for the 3 weeks. It comes on when she laughs, exercises, or yawns. Her mother says it has a "whooping" sound. Nothing makes it better. It occurs day and night and is interrupting her sleep and her studies. The patient reports that last month "I had a cold with runny nose, mild cough, general unwell feeling, and maybe a slight fever". An outbreak of *Bordetella pertussis* infection has been reported in the region.

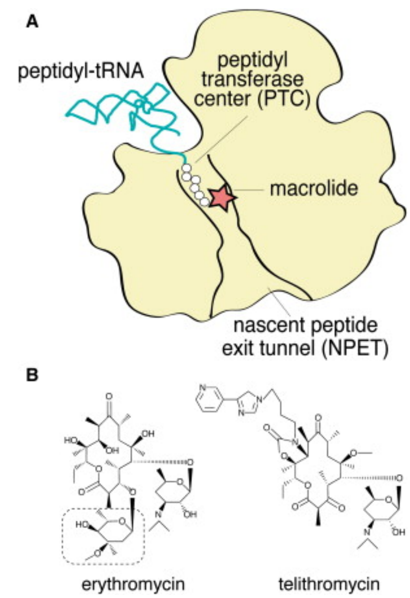
Nasopharyngeal and post-pharynx swabs are obtained for bacterial culture and PCR. The patient is given a clinical diagnosis of whooping cough. An e-prescription is submitted for empiric treatment with a drug that is widely distributed in the body, including lung tissue and macrocytes. The drug is excreted in the feces and has a half-life of ~70 hours, due to sequestration of the drug in tissues. The course of therapy is 5 days with a loading dose on day 1. Due to the long half-life, 5 days of therapy provides a therapeutic effect for up to 10 days.

Which one of the following best describes this drug's mechanism of action?

A. Binds 30S subunit → inhibits formation of the initiation complex and causes misreading of the mRNA
B. Binds 30S subunit → prevents binding of charged aminoacyl tRNA
C. Binds 50S subunit → prevents the new protein from exiting the ribosome*
D. Binds 50S subunit → inhibits peptidyl transferase activity and peptide bond formation
E. Binds 50S subunit → prevents formation of the ribosomal-fMet-tRNA complex

Hint: *B. pertussis* treatment is on p16 of the Notes Handout.

I recommend you go to YouTube and listen to Whooping Cough Sound.



Rationale: C.

The big clues – "whooping" cough and *B. pertussis*

The described drug is the first-line treatment: macrolide, azithromycin.

1- Macrolides reversibly bind the 23S rRNA of the 50S subunit near the peptidyl transferase center.

2- Chain elongation (transpeptidation) is prevented by blocking the polypeptide exit tunnel of the large subunit – the tunnel through which the newly synthesized peptide leaves the ribosome. (Some refer to macrolides as "tunnel plugs".)

3- As a result, the unfinished peptide dissociates from the ribosome.

Incorrect options:

- Binds 30S subunit → causes misreading of the mRNA: Aminoglycosides
- Binds 30S subunit → prevents binding of charged aminoacyl tRNA: Tetracyclines
- Binds 50S subunit → inhibits peptidyl transferase activity, thereby preventing peptide bond formation: Chloramphenicol

  Clindamycin also binds near the peptidyl transferase center, disrupts the proper positioning the A-site and P-site, which prevents the formation of a peptide bond between the incoming amino acid and the last amino acid on the peptide chain.
- Binds 50S subunit → prevents formation of the ribosomal-fMet-tRNA complex: Oxazolidinones linezolid and tedizolid

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13. Imagine you wish to treat a patient for pertussis (whooping cough) and the supply of the long-acting antibiotic described in the previous question is in shortage due to an environmental disaster at the manufacturing facility.

What is an alternative agent from the same class of antibiotics that could be given?

Short answer

The rationale is under the MACROLIDES table.

MACROLIDES: WHAT YOU NEED TO KNOW

Erythromcyin, Clarithromycin, Azithromycin

Takeaways: PK differences between the drugs, MOA, G+, atypical, some G– respiratory pathogens; widespread resistance emerging; intrinsic resistance (see below); Tx of respiratory tract infections, H. pylori, MAC; Toxicities especially QT interval prolongation.

PK:

- All: Oral; extensive distribution; poor CNS penetration; biliary/fecal excretion
- Erythro-: poor unreliable oral bioavailability
- IV admin: erythromycin and azithromycin
- Hepatic metabolism and biliary excretion.
- Erythromycin and clarithromycin: CYP3A4 metabolism and inhibition
- Azithromycin minor CYP3A4 substrate and biliary excretion of unchanged drug
- Clarithromcyin: Dose adjustment for renal insufficiency
- Half-life: Erythro- short; Clarithro- longer (BID); Azithro – very long

PK-PD profile:

Clarithro-/ Azithro-: mixed: AUC/MIC; moderate PAE and time-dependent T>MIC

Erythro: not listed

Mechanism

Tunnel plug: Macrolides bind 50S ribosome near the peptidyl transferase center and block the exit tunnel of the growing polypeptide chain, which stops protein synthesis.

Spectrum of activity

- Gram-positive cocci
- Gram-negative*, Haemophilus influenzae, Moraxella catarrhalis*

Resistant pathogens:

- MRSA – widespread resistance (erm gene, mrsA gene)
- S. pneumoniae resistance widespread (ermB gene)
- *Neisseria gonorrhoeae*: target modification and efflux pumps – Azithromycin (and the others) – a serious and growing problem
- **Intrinsic resistance:** *Staphylococcus epidermidis, Enterococcus*, Enterobacterales, *Pseudomonas, B. fragilis, C. difficile*

Resistance mechanisms:

1) Ribosomal modification: Most common mechanism.

- rRNA binding site methylation encoded by erm genes. G+ resistance.
- MLS_B phenotype: Cross-resistance macrolides, clindamycin, streptogramin B (quinupristin)

2) Efflux pumps: Mef and Msr genes, and othersz;: G+ and G– pathogens (*H. influenzae, M. catarrhalis*)

3) Enzymatic drug modification: G– mainly: esterases, phosphotransferases

4) Ribosome protection: discovered in a wide range of pathogens

Therapeutic uses

First-line for:

- Atypcial pneumonia
- Pertussis (whooping cough)
- Chlamydia uncomplicated genital infections
- Pneumocystis pneumonia – an AIDS-defining fungal infection that is an indicator of advanced HIV.
- *H. pylori* eradication (peptic ulcer): Clarithromycin
- MAC: First line – azithro- or clarithro-

Alternative: Mild S. pyogenes skin/soft tissue infection

TOXICITY

- **Infants:** Displaces bilirubin on albumin binding sites – may cause kernicterus in young infants and potential for drug interactions
- **G6PD deficiency:** Dose-related potential for hemolytic anemia in patients with G6PD deficiency
- **Hypersensitivity reactions** – skin rash and systemic symptoms – fever, SJS/TEN, hepatoxicity, blood dyscrasia
- **Crystalluria nephrotoxicity** (acidic pH)

Toxicities

- Gi upset, esp. erythro- due to activation of motilin receptors
- QT interval prolongation
- Heptotoxicity

Drug interactions

- Erythro- / Clarithro- High potential for drug interactions

Rationale: The other macrolides, erythromycin (QID – q6h x14 days) or clarithromycin (q12h x7 days), would also be effective against *Bordetella pertussis* infection. TMP-SMX (q12h x14 days) is an alternative non-macrolide.

- **Practice point:** Drug shortages are common. You must be prepared with a knowledge of effective alternatives when necessary.

Causes include shortages of raw materials, manufacturing and quality problems, delays in availability, and discontinuations. Information on drug shortages can be found at https://www.accessdata.fda.gov/scripts/drugshortages/

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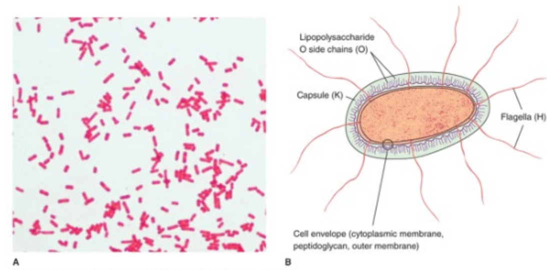
14. A 79-year-old febrile woman with advanced Alzheimer disease, diabetes mellitus type 2 (T2DM or T2D), and recurrent acute urinary tract infections (UTIs) is admitted for treatment of a complicated UTI. She has been receiving cephalexin for UTI prophylaxis for 1 month. Previous UTIs were repeatedly treated with trimethoprim-sulfamethoxazole (TMP-SMX), nitrofurantoin, cephalexin, ciprofloxacin, and ceftriaxone. She has a history of allergy to penicillin (anaphylaxis). The patient resides in the Memory Care unit of a skilled nursing facility.



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

A: Gram-stain of E. coli. Original magnification × 1000. (Courtesy of H Reyes.) B: Antigenic structure of Enterobacteriaceae.

Citation: Chapter 15 Enteric Gram-Negative Rods (Enterobacteriaceae), Riedel S, Hobden JA, Miller S, Morse SA, Mietzner TA, Detrick B, Mitchell TG, Sakanari JA, Hotez P, Mejia R. *Jawetz, Melnick, & Adelberg's Medical Microbiology*, 28e; 2019. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2629&sectionid=217770983 Accessed: November 03, 2020

Symptoms and signs are consistent with urosepsis. Dipstick urinalysis is positive for leukocyte esterase and nitrites, suggestive of infection with Gram-negative bacilli. Urine Gram stain shows Gram-negative rods. Blood and urine cultures are pending.

The patient is placed on empiric intravenous therapy for coverage of *E. coli*, *Klebsiella* spp*,* and *Proteus mirabilis*. The selected drug is highly effective against most aerobic Gram-negative bacteria. Despite the short 2-hour half-life, once daily IV administration is effective. Because of potential drug toxicity, renal and eighth cranial (vestibulocochlear) nerve function will be closely monitored.

Definitions:

- Complicated UTI refers to a UTI with features suggesting the infection extends beyond the bladder, such as fever, chills or rigors, significant fatigue or malaise, and flank pain.
- Urine nitrite test has a very high sensitivity (>90%). A positive test reflects the presence of Enterobacterales, which convert nitrate normally present in urine to nitrite.

Which one of the following antibiotics fits the activity profile and is likely to be effective in this patient?

A. Ampicillin-sulbactam
B. Cefotaxime
C. Doxycycline
D. Gentamicin*
E. Levofloxacin
F. Neomycin
G. Sulfamethoxazole

Rationale: D. gentamicin

Thinking it through:

a) A febrile patient presents with a serious infection requiring hospitalization for treatment.

b) Urinalysis and Gram stain indicate the presence of Gram-negative bacilli. *E. coli* is the most frequent cause of complicated acute UTIs.
   Other uropathogens include *Proteus* spp, *Klebsiella* spp, *Pseudomonas*, and Gram-positive bacteria, enterococci and staphylococci. The prevalence of a particular pathogen depends on the host.
c) The patient has been given several antibiotic regimens from different classes for prophylaxis and treatment of recurrent UTIs, suggesting high risk of multidrug resistant infection.
d) Empiric therapy should consist of antibiotics likely to be effective.
e) The patient is allergic to penicillin (anaphylaxis) so a beta-lactam antibiotic is avoided.
f) The patient will be closely monitored for toxicity to the kidneys and for hearing and vestibular function. The aminoglycosides class has a high potential for these toxicities.
g) Gentamicin is highly active against Gram-negative bacteria and the patient has not previously been exposed to an aminoglycoside. The infectious organisms are likely to be sensitive to the new treatment regimen.

➢ **Practice points:** The antibiotic regimen should be changed to a less toxic drug when the culture and sensitivity results are available – definitive (= directed) therapy.
Typically, aminoglycosides are not used for more than 48 hours because of toxicity, unless an aminoglycoside is appropriate directed therapy for the patient's infection.

Change to a less toxic drug when culture and sensitivity results are available.

Jawetz, Melnick, and Adelberg's Figure 15-1: A: Gram stain of E. coli. Original magnification ×1000. (Courtesy of H Reyes.) B: Antigenic structure of Enterobacteriaceae.

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15. A 6-month-old girl is brought to the local emergency department by her mother, who reports that the child has a fever, irritability, sleepiness, doesn't want to eat, and has not vomited. Her family immigrated from Haiti one year ago. Family history reveals that the patient's mother suffered an untoward reaction when she was treated for a urinary tract infection with "a sulfa drug" 6 years ago. The mother was told at the time that she has a G6PD deficiency.

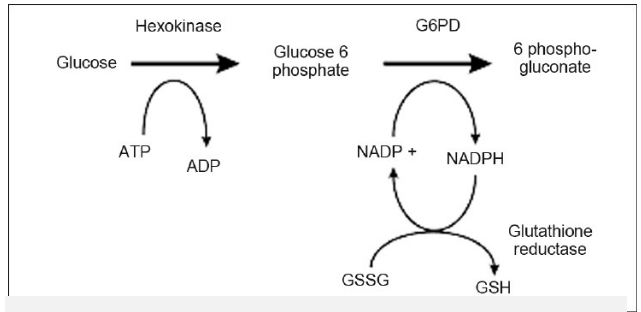
Urinalysis obtained by catheterization is positive for leukocyte esterases and urinary nitrites. Gram stain reveals Gram-negative rods. A urine sample is sent to the microbiology lab for culture.

Based on clinical presentation, the patient is given a diagnosis of uncomplicated urinary tract infection. In view of the family history, trimethoprim-sulfamethoxazole is contraindicated because its potential toxicity in this patient.

What is the adverse drug reaction?

A. Hemolytic anemia*
B. Neutropenia
C. Sickle cell anemia

D. Thalassemia
E. Thrombocytopenia



G6PD maintains the body's supply of reduced NADPH. G6PD catalyzes the reduction of NADP+ to the co-enzyme nicotinamide adenine dinucleotide phosphate, NADPH.

Definition: G6PD: glucose-6-phosphate dehydrogenase is an enzyme crucial for preventing cellular damage by reactive oxygen species.

Rationale: A. Hemolytic anemia

Students, this is an important mechanism for you to understand.

Mechanism: G6PD is a cytosolic enzyme that participates in the pentose phosphate pathway, which supplies energy to cells, including red blood cells.

G6PD maintains levels of NADPH, which in turn maintains levels of reduced glutathione (GSH).

Glutathione helps protect cells from oxidative damage by compounds like hydrogen peroxide (H2O2) and superoxide anion ($O2^{-}$).

Red blood cells (RBCs) are especially sensitive to oxidative stress. The G6PD pathway is the only source of NADPH in red blood cells. RBCs have relative high levels of GSH to protect them against oxidative injury. Individuals with G6PD deficiency are susceptible to hemolytic anemia when exposed to drugs with a high oxidant potential.

Heritability: G6PD deficiency is an X-linked disorder encompassing a spectrum of hemolytic disorders. Affected people, including females, are often asymptomatic. Many have episodic anemia. Some patients have chronic hemolysis. Susceptibility is related to the inherited variant.

Incorrect options: None of the conditions in the other options is directly exacerbated by sulfonamides, trimethoprim, or quinolones.

- Neutropenia: Drugs that harm the bone marrow may result in neutropenia.
- Sickle cell anemia: A heritable disease due to point mutations in the beta globin gene of hemoglobin
- Thalassemia: A group of heritable disorders due to variants in one or more globin genes.
- Thrombocytopenia: Drugs that harm the bone marrow may result in thrombocytopenia.

Note: Severe hypersensitivity reactions to sulfonamide therapy can result in a constellation of effects, including agranulocytosis or aplastic anemia. This effect is not triggered by deficiency in G6PD.

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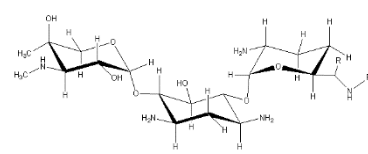
16. How do the structure and physicochemical properties of the aminoglycosides contribute to their pharmacokinetic properties:



· parenteral only · distributes in extracellular fluid · does not cross BBB
· renal elimination of unchanged drug by glomerular filtration
· t½ 2-3 hours, adults, normal renal function · t½ 24-48 hours, renal impairment

Short answer.

Rationale: Aminoglycosides are small, polar molecules.

Leslie Goldstein, Pharm.D.
Updated September 2025

Aminoglycosides are polycationic amino sugars. Being polar molecules, they are highly hydrophilic.

- They do not diffuse through lipid bilayers, thus they do not diffuse through physiologic barriers – they are not absorbed from the GI tract and they remain in extracellular fluid when administered parenterally.
- They are small soluble molecules disbursed in serum. Thus, they are filtered with the plasma through the glomerulus.
- They are not metabolized. Thus, they achieve high concentrations of active drug in urine.

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17. An adult patient with a history of dysphagia is admitted for the management of a lung abscess secondary to aspiration pneumonia. The electronic medical record (EMR) indicates an allergy to penicillin (anaphylaxis). The patient is treated with an intravenous antibiotic from one of the following classes because of its effectiveness against the most frequent causative organisms, microaerophilic *Streptococci* species and strict anaerobic species, which are components of the normal flora of the respiratory tract.

Which one?

A. Aminoglycosides
B. Lincosamides*
C. Macrolides
D. Oxazolidinones
E. Streptogramins
F. Tetracyclines

Hints:

Think about which antibiotic in this series of lectures is useful against Gram-positive infections and anaerobic infections "above the waist".

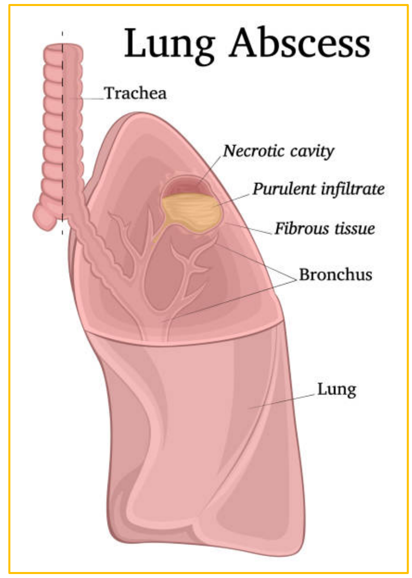

Illustration of a sectioned lung depicting a purulent-necrotic cavity (abscess) https://www.istockphoto.com/vector/illustration-of-lung-abscess-gm1124712879-295366028

Lung Abscess: A circumscribed area of pus or necrosis in the pulmonary parenchyma, commonly caused by microaerophilic *Streptococci* species and anaerobes *Peptostreptococcus, Prevotella,* and *Fusobacterium*. Students, you don't need to know the names of these anaerobic bugs. "Anaerobic" is the takeaway.

Rationale: B. Lincosamides

Clindamycin is the only drug in the lincosamides class marketed in the U.S.

❖ Pathophysiology of a lung abscess is typically:

Hypoxic environment favors survival of anaerobes, which thrive in low-oxygen, low pH environments.

Acidic environment favors anaerobic metabolism – fermentation pathways – which produce lactic acid.

Acidic environment impairs immune cell function.

Empiric therapy for the treatment of lung abscess should include antibacterial agents that penetrate lung parenchyma and are effective against potential infective pathogens.

- **Practice points:** Traditionally, clindamycin has been used for the treatment of anaerobic infections “above the waist”. It is a treatment option depending on the infective organism and other factors.

Clindamycin’s spectrum of activity is easy to remember:

Effective: Streptococci, staphylococci, and anaerobes (both Gram-positive and Gram-negative).

- Clindamycin is not recommended for treatment of the enteric anaerobic Gram-negative rod *Bacteroides fragilis* (“below the waist”) due a high rate of resistance.

Not effective: Clindamycin has ***no activity*** against aerobic Gram-negative bacilli (Enterobacterales, *Pseudomonas*, etc).

Incorrect options: The drugs in the other options do not fit the criteria for empiric therapy of lung empyema. (See the table in the SUMMARY at the end of this document.)

<table>
<tr><td colspan="2">CLINDAMYCIN: WHAT YOU NEED TO KNOW
Takeaways: PK, MOA; streptococcal, staphylococcal, and anaerobic infections; resistance by MLS_B methylation of ribosome; D-test for detection of $iMLS_B$; higher potential for C. difficile infection compared to other antibiotics</td></tr>
<tr><td>PK
· Oral, IV, IM; good penetration into abscesses, bone, and soft tissue; CYP3A4; small amounts excreted in urine/feces
Mechanism of action
· Passive diffusion across membranes
· Binds 23S rRNA on 50S subunit → interferes with proper positioning of the A-site and P-site, which inhibits transpeptidation → chain elongation stops
· Also blocks the exit tunnel
Spectrum: Bacteriostatic
· Streptococci, Staphylococci, anaerobes
· BUT widespread and increasing resistance
· Intrinsic resistance: Enterococci, aerobic G- (Enterobacterales, Pseudomonas)</td><td>Resistance mechanisms
· Strep and Staph: Ribosomal methylation by MLS_B (erm gene-encoded methylase). Inducible MLS_B is prevalent in MRSA. (D-test advised)
Therapeutic uses (currently)
· Primarily used for serious bacterial infections, especially in patients with penicillin allergy.
· Effective against susceptible MRSA and anaerobic infections.
TOXICITY
· C. difficile associated colitis: Clindamycin is frequently implicated.
· Painful IM injection
· Hypersensitivity reactions</td></tr>
</table>

18. A 64-year-old man with history of insulin-dependent diabetes mellitus (IDDM) presents to the ED with fever, chills, cough with purulent sputum, and shortness of breath with use of accessory muscles for breathing. He reports allergies (anaphylaxis) to penicillin and ceftriaxone.

- Physical examination, clinical laboratory tests and chest radiograph results are consistent with left lobar pneumonia. Sputum and blood samples are obtained for culture.
- Left lower lobe opacity is noted on chest radiograph.
- The patient is admitted for empiric treatment of pneumococcal pneumonia. He is at low risk for MRSA and *Pseudomonas* infection.

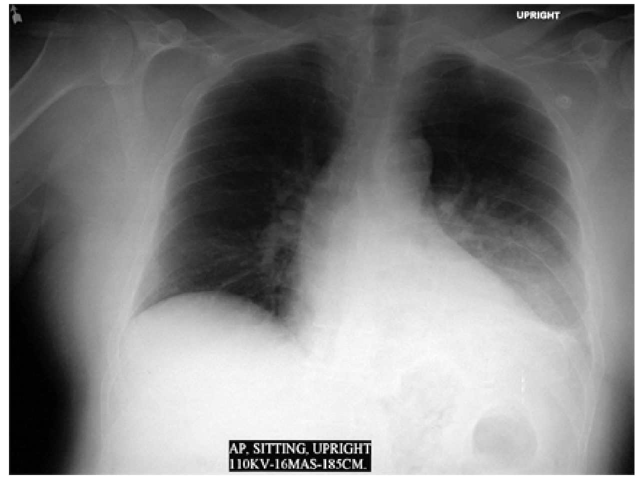

Figure: UpToDate Graphic 6568 version 4.0: Pneumococcal pneumonia chest radiograph.
64-year-old male with insulin-dependent diabetes mellitus. He was admitted with bacteremic pneumococcal pneumonia. Note the left lower lobe opacity.

i) Which one of the following options is bactericidal against susceptible strains of the infectious pathogen and is not contraindicated based on this patient's allergy history?

A. Cefotaxime
B. Chloramphenicol
C. Ciprofloxacin
D. Levofloxacin*
E. Linezolid
F. Piperacillin-tazobactam

ii) What alterative agent from the same antibiotics class would also be effective?

Short answer

Hint: Think about which class of drugs should be avoided. Then, think about which drugs are effective in treating the likely infective pathogen.

Rationale: Levofloxacin, moxifloxacin, and delafloxacin have excellent activity against susceptible strains of the common infectious respiratory pathogens of community-acquired pneumonia (CAP): respiratory pathogens *S. pneumoniae, Haemophilus influenzae, Moraxella catarrhalis,* and the bacteria that cause "atypical" pneumonia, *Mycoplasma pneumoniae*, *Chlamydophila pneumoniae*, and *Legionella pneumophila*.

There is no cross-reactivity between the fluoroquinolones (FQs) in beta-lactam allergic individuals.

➢ **Practice point recall:** Piperacillin-tazobactam is useful for the treatment of ventilator-associated HAP because of its broad spectrum of activity, including antipseudomonal activity.

❖ **Microbiology/Pharmacology recall:**

Bacteria that cause atypical pneumonia:

– *Mycoplasmas* are of extremely small size and lack a cell wall.

Beta-lactam antibiotics target the transpeptidases involved in the synthesis of the peptidoglycan cell wall. They are, therefore, ineffective against mycoplasma.

- *Chlamydia* spp are obligate intracellular organisms and their cell wall contains a minimal and modified form of peptidoglycan in small amounts.

  Beta-lactam antibiotics are, therefore, ineffective because they do not penetrate the plasma membrane.
- *Legionella* are Gram-negative obligate intracellular organisms. Beta-lactam antibiotics are unable to penetrate the macrophages where Legionella thrive inside "Legionella-containing vacuoles". Beta lactams are, therefore, ineffective.

❖ Only drugs able to cross cell membranes and get to the intracellular bacterial hidey-holes are effective against intracellular organisms.

Incorrect options:

- A. Cefotaxime is structurally similar to ceftriaxone and cross-reactive in patients reporting allergy to ceftriaxone, other third generation cephalosporins, and cefepime.
- B. Chloramphenicol: Broad spectrum agent with limited use due to its potentially life-threatening toxicities; it is bacteriostatic.
- C. Ciprofloxacin: Second-generation FQ with relatively low activity against pneumococci, although it is useful for Gram-negative respiratory pathogens.
- E. Linezolid: Good activity against Gram-positive bacteria but not Gram-negative respiratory pathogens. Linezolid may be used in hospital-acquired pneumonia (HAP) for coverage of MRSA.
- F. Piperacillin-tazobactam: Contraindicated due to anaphylactic reactions to penicillin; beta-lactam allergy reported.

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19. Which one of the following best represents the mechanism of action of the drug selected for the patient in the case vignette above?

The drug:

A. Causes production of aberrant nonfunctional proteins by misreading mRNA and incorporating incorrect amino acids.

B. Inhibits the formation of a cofactor that participates in the synthesis of purines and pyrimidines

C. Introduces DNA strand breaks by preventing the relaxation of DNA supercoils*

D. Prevents translation of mRNA to form proteins

E. Triggering autolysin activity and osmotic lysis

Rationale: The selected drug is levofloxacin.

Levofloxacin, moxifloxacin, and delafloxacin are "respiratory fluoroquinolones" because they are active against *S. pneumoniae*.

Ciprofloxacin exhibits poor and inconsistent activity in vitro and in vivo.

Delafloxacin has good Gram-positive activity, including MRSA and good activity against many Gram-negative bacteria. It is very expensive.

Mechanism of action: FQs inhibit DNA synthesis by blocking the actions of DNA gyrase (bacterial DNA topoisomerase II) and/or bacterial topoisomerase IV, resulting in strand breaks and death of the microorganism.

All quinolone/fluoroquinolones have activity at both target sites. Most have higher affinity for one or the other. Delafloxacin has balanced affinity for both targets.

- DNA gyrase appears to be the primary target in Gram-negative bacteria.
- Topoisomerase IV appears to be the target in many Gram-positive bacteria, including *S. pneumoniae*.

The different affinities for the bacterial topoisomerases may explain the differences in the different FQs' spectrums of action. (Katzung, 2024)

Molecular biology: Unwinding of the DNA generates supercoils. DNA gyrase (a type of topo II) introduces double strand negative supercoils, which relaxes the positive supercoils by nicking and resealing the bacterial DNA during replication.

Helpful video: https://www.youtube.com/watch?v=IkKZ_gxAOXI

Topoisomerase IV separates newly synthesized daughter cells (called decatenation) by binding the intertwined segments, creating a double-strand break, passing the unbroken strand through, and resealing.

The DNA gyrase and topo IV process are explained on page 3 of the Notes.

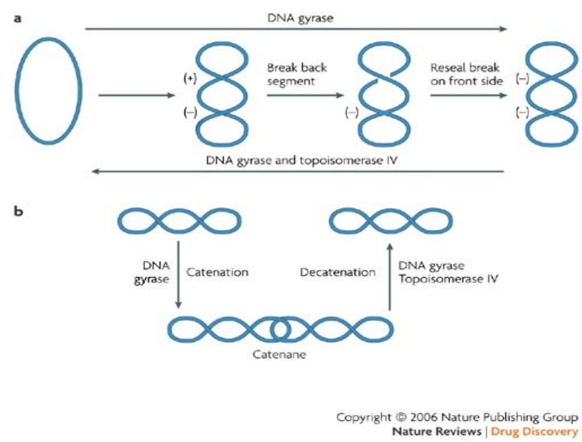

Figure: Silver Nature Reviews Drug Discovery 6, 41–55 (January 2007) | doi:10.1038 / nrd2202 http://www.nature.com/nrd/journal/v6/n1/fig_tab/nrd2202_F3.html

Incorrect options:

- Aminoglycosides cause production of aberrant nonfunctional proteins by misreading mRNA and incorporating incorrect amino acids.
- Sulfonamides and trimethoprim inhibit the formation of a cofactor, tetrahydrofolate, that participates in the synthesis of purines and pyrimidines.
- Protein synthesis inhibitors prevent translation of mRNA to form new proteins.
- Beta-lactam antibiotics block the cross-linking of the peptidoglycan subunits, which triggers autolysin activity and osmotic lysis of the bacterial cell.

FLUOROQUINOLONES: WHAT YOU NEED TO KNOW **Ciprofloxacin, Levofloxacin, Moxifloxacin, Delafloxacin** Takeaways: excellent tissue penetration, renal excretion; AUC/MIC, inhibition of DNA gyrase/topo IV; numerous potentially serious toxicities	
PK · Oral, IV, IM; Excellent bioavailability · Excellent penetration into tissues – lung, skin/soft tissue, bone/joints, kidneys, prostate, CSF (levo, moxi, dela), neutrophils and macrophages. · Excreted unchanged in urine · Moxi- some hepatic metabolism with low amounts in urine **PK-PD profile:** Mixed AUC/MIC with prolonged PAE and Peak/MIC **Mechanism of action** · DNA gyrase and topo IV inhibition interrupts DNA replication, transcription, repair, and replication · G– target mainly DNA gyrase; G+ mainly topo IV · Dela- targets both about equally **Spectrum:** Bactericidal · Broad spectrum, G+ and G– bacteria including *Pseudomonas* · Atypical bacteria (*Mycoplasma, Chlamydia, Legionella*) · Only dela- is active against MRSA · Moxi- and dela- have activity against most anaerobes (not cipro- or levo-)	**Resistance mechanisms** · Chromosomal: Target modification common and high-level resistance – mutations in gyrA (DNA gyrase) and parC (topo IV) · Chromosomal: Efflux pumps and decreased expression of porins · Plasmid-mediated transfer of Qnr proteins the bind and protect DNA gyrase and topo IV FQ binding targets · Cross-resistance: Resistance to one confers resistance to the others EXCEPT to dela-. · **High rates of resistance** in *E. coli, Klebsiella pneumoniae, Pseudomonas aeruginosa,* and *Acinetobacter baumannii* · Intrinsic resistance: Enterococci and anaerobes **Therapeutic uses (currently)** · Broad antibacterial spectrum · Wide variety of infections in a wide variety of tissues. · Cipro- has weak activity against *S. pneumoniae*. · Only dela- is effective against MRSA

FQs ARE ASSOCIATED WITH NUMEROUS SERIOUS TOXICITIES	
· *C. difficile* associated diarrhea · Tendonitis/tendon rupture · Peripheral neuropathy · QT prolongation · Aaortic aneurysm/dissection	· Hepatotoxicity · Nephrotoxicity · Myasthenia gravis exacerbation · Psychiatric effects: abnormal dreams, agitation, confusion, dizziness, headache
Not recommended for: · Children <18 years old – risk of joint damage · Pregnant or breastfeeding women · Patients with myasthenia gravis · Patients with certain heart rhythm disorders	Monitor for signs/symptoms of toxicities: · Tendon pain, swelling inflammation, rupture · Numbness, tingling, burning or sharp pain, which often starts in hands or feet · Blood glucose (hypo- or hyperglycemia) · Rash / signs and symptoms of hypersensitivity reaction

20.i) Which one of the following *best* represents the PK-PD profile of the selected drug in the case vignette above?

A. The antimicrobial concentration remains above the MIC for most of the dosing interval.

B. A high peak concentration is achieved with a prolonged persistent antimicrobial effect.

C. The total drug concentration (exposure) over time in relation to the MIC and moderate to prolonged persistent antimicrobial effect.

D. Mixed PK-PD profile, total drug concentration over time and high peak concentration, are predictors of efficacy.*

E. Mixed PK-PD profile, total drug concentration over time and duration above the MIC are predictors of efficacy.

ii) What information do the pharmacokinetics-pharmacodynamics profiles provide?

Rationale: D. Fluroquinolones exhibit a mixed AUC/MIC ratio and peak/MIC ratio profile

PK-PD profiles *are used to determine optimal dosing*.

AUC_{24}/MIC: The primary predictor, most consistently correlated with efficacy

- Efficacy and Dosing: The ideal dosing regimen for these antibiotics maximizes the **amount** of drug received in a 24-hour period, which is determined at steady state.
- AUC_{24}/MIC ratio has positive predictive value for *S. pneumoniae, Mycobacterium tuberculosis,* and *Pseudomonas aeruginosa.*

C_{max}/MIC Contributes to overall pharmacodynamic activity

- Predictive of clinical and bacteriological outcomes, especially against Gram-negative bacteria
- **PAE:** Moderate post-antibiotic effect – generally, 1.5-3 hours after FQ concentration falls below the MIC, depending on the targeted pathogen.

Killing effect: Bactericidal

➢ **Practice point:** The AUC_{24}/MIC values that correlate with increased probability of successful microbial and clinical outcome vary among the different organisms and drugs.

For example, predictive values in adult patients

Gram-negative infections: Ciprofloxacin AUC_{24h}/MIC $\geq$125
Streptococcus pneumoniae: Levofloxacin AUC_{24h}/MIC $\geq$30
Mycobacterium tuberculosis: Moxifloxacin AUC_{24h}/MIC = 23

Students, what you need to know is **the concept**:
The PK-PD profile is specific for specific organisms and can help guide dosing for the individual patient.
You don't need to memorize the numbers.

21. What are the potential toxicities and the mechanisms associated with fetal adverse effects if the mother is taking a drug from the tetracyclines group for treatment of an infection?

A. Deformity of the long bones and bone fragility

B. Hematopoietic disturbances and anemia

C. Induction of nephrogenic diabetes and polyuria

D. Kidney stones and obstruction of urine flow

E. Tooth enamel dysplasia and bone growth disruption*

Rationale: E. Tooth enamel and bone growth disruption

Calcium and other di- and trivalent cations chelate tetracyclines.

Complexing with calcium in growing bones and tooth enamel can potentially interfere with the growth of long bones and stain the permanent teeth. These effects are more likely to occur following long-term or repeated exposure.

The actions of tetracyclines on developing bone appear to be reversible when the drug is stopped.

Tooth discoloration is permanent.

Exception: Doxycycline has reduced ability to chelate calcium compared to other tetracyclines and is less likely to cause permanent tooth discoloration.

Incorrect options:

- A. Vitamin D is an essential nutrient important for calcium homeostasis. Severe vitamin D deficiency results in architectural disruption of the bones. It causes rickets in infants and children and osteomalacia in all age groups.
- B. Tetracyclines are not associated with blood dyscrasias. This toxicity is associated with chloramphenicol use.
- C. Demeclocycline can induce nephrogenic diabetes insipidus by reducing responsiveness in the collecting tubules to antidiuretic hormone. This is a unique feature of demeclocycline. The drug is not used for its antibacterial effects.
- D. Tetracyclines do not cause kidney stones.

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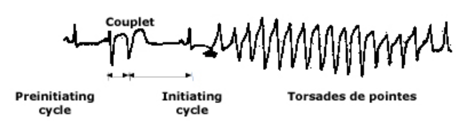
22. A 22-year-old person with a history of schizophrenia treated with ziprasidone (antipsychotic medication) is diagnosed with pneumonia. NKDA. This antipsychotic medication has been associated with QT interval prolongation. Several antimicrobials also increase QT interval. Taking them at the same time increases the risk of ventricular arrhythmia and death.

Which one of the following antimicrobials for empiric treatment of this patient's lung infection is unlikely to increase this risk?

A. Azithromycin
B. Ampicillin-sulbactam*
C. Lefamulin
D. Moxifloxacin
E. Voriconazole

Single lead electrocardiogram (ECG) showing torsades de pointes



The electrocardiographic rhythm strip shows torsades de pointes, a polymorphic ventricular tachycardia associated with QT prolongation. There is a short, preinitiating RR interval due to a ventricular couplet, which is followed by a long, initiating cycle resulting from the compensatory pause after the couplet.

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Rationale: The beta-lactams do not prolong the QT interval.

Important!

- **Practice point:** *Acquired long QT syndrome usually results from drug therapy*. Patient-specific and medication-related factors can enhance it potentially leading to torsades de pointes (TdP), which can lead to ventricular arrhythmia and sudden cardiac death.

· Patient-specific factors include:
  - Electrolyte imbalances
  - Structural and ischemic heart disease
  - Bradyarrhythmias, and
  - *Concomitant drugs that cause QT interval prolongation – a long list*

· Drug-related factors that increase the risk of the patient developing TdP include:
  - High doses, such as I.V., which rapidly achieve high drug concentrations, vs oral
  - Drug interactions that increase concentrations of the QT prolonging drug, such as inhibition of metabolism of renal excretion
  - Taking two or more QT prolonging drugs concomitantly.

Know the mechanisms of the drug-related causes of TdP.

Examples include:
  - Some antimicrobials – macrolides, FQs, lefamulin (a pleuromutilin protein synthesis inhibitor antibiotic), and azole antifungal agents (voriconazole in the above options)
  - Some antiarrhythmic agents
  - Some antipsychotic agents
  - Some antidepressants
  - Some synthetic opioids (methadone)
  - Some antiemetics

THE TAKEAWAY
Torsades de pointes ventricular arrhythmias can be deadly. Many drugs and patient factors increase the risk.
Be sure to check the literature when considering drug therapy.

Figure: UpToDate Graphic 73827 v. 4.0: Single lead electrocardiogram (ECG) showing torsades de pointes

The electrocardiographic rhythm strip shows torsades de pointes, a polymorphic ventricular tachycardia associated with QT prolongation. There is a short, preinitiating RR interval due to a ventricular couplet, which is followed by a long, initiating cycle resulting from the compensatory pause after the couplet.

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23. The local community hospital has seen increasing resistance to doxycycline.

i) PCR analysis shows bacterial expression of tetM gene, which encodes a(an):

A. Acetylase
B. Efflux pump
C. Hydrolase
D. Methylase
E. Ribosomal protection protein*

ii) What is the potential for cross-resistance due to this resistance mechanism?

Hint: I think you have identified the search term to help you find the answer in the Notes. Try to answer the question before looking up the answer. This practice will help you remember it.

Rationale: E. ribosomal protection protein

The ribosoMal protection protein encoded by the tetM gene expresses ribosomal protection proteins, reducing affinity of the tetracycline binding site for the drugs.

This mutation confers streptococcus, staphylococcus, and enterococcus resistance to tetracycline, doxycycline, and minocycline,

But not to the newer agents – tigecycline, eravacycline, or omadacycline

Mnemonics:

MNEMONICS

TetM – M is in ribosoMe → Gram-positive resistance the older tetracyclines group

K almost looks like X – efflux

K sounds like C – Cocci (*Staphylococcus aureus* is a Gram-positive coccus)

Tet(AE) gene → efflux pump → Gram-negative bacteria → resistance to older tetracyclines

A=aerobic; E=enteric = Enterobacterales, E-Efflux

The newer agents in the tetracycline class, tigecycline, eravacycline, omadacycline, and sarecycline have activity against tetracycline-resistant strains because their action is not inhibited by ribosomal protection proteins and they are not substrates of many of the bacterial efflux pumps.

Incorrect options:

– Acetylase: Aminoglycosides and Chloramphenicol. Aminoglycosides may be inactivated by adenylation and phosphorylation also.

tetK gene

– Efflux pump → tetK gene → S. aureus → tetracycline but not the rest

  tetK expression by tetracycline-resistant strains of *S. aureus* encodes an efflux pump that extrudes the drug from the bacterial cell confers resistance to tetracycline, *but not to* doxycycline, minocycline, tigecycline, eravacycline, or omadacycline.

– Hydrolase: Beta-lactamase – beta-lactam antibiotics

– Methylase: Macrolides

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24. A study on fluoroquinolone resistance testing was performed in an ongoing Sexually Transmitted Infections surveillance program in different locations in Kenya between 2013-2017. Antimicrobial Susceptibility Tests were performed in 84 gonococcal isolates; 22 isolates were selected for sequencing. The isolates were subcultured and whole genomes were sequenced. 20 of the isolates were found to be resistant to ciprofloxacin. Gene mutations were identified that conferred reduced affinity of the microbial “receptor” for the fluoroquinolones.

Which of the following proteins was altered due to this gene mutation?

A. Dihydropteroate synthase
B. Enzymes that protect the ribosome acceptor site
C. Lanosterol demethylase
D. PBP2a
E. Topoisomerases*

Rationale: E. Topoisomerases

The mutations in the genes expressing the FQ binding areas in bacterial DNA gyrase and topoisomerase IV resulted in reduced affinity of the enzymes for the drugs, thereby conferring resistance to the FQs.

Incorrect: All the other options are targets of other classes of antimicrobials.

- A. Dihydropteroate synthase: Sulfonamides
- B. Enzymes that protect the ribosome acceptor site: Protein synthesis inhibitors, notably tetracyclines and macrolides
- C. Lanosterol demethylase: Azole antifungals (Students: You can rule this one out by, "I never heard of this one."
- D. PBP2a: Penicillin binding protein (transpeptidase) expressed by *S. aureus* with low affinity for beta-lactams

Resources: Kivata MW, Mbuchi M, Eyase FL, Mancuso JD, et.al. gyrA and parC mutations in fluoroquinolone resistant *Neisseria gonorrhoeae* isolates from Kenya. BMC Microbiology (2019) 19(76) Open Access. https://bmcmicrobiol.biomedcentral.com/articles/10.1186/s12866-019-1439-1

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25. A 22-year-old woman presents to her PCP with complaints of a nonproductive cough, fever, "bad" headache, muscle aches, fatigue, and photophobia for 3 days. Acetaminophen reduced the fever for 4 to 5 hours but did not relieve the headache or muscle aches. She returned from a visit to her parents' sheep farm in the South of France 15 days ago. NKDA. Her temperature is 39.1°C (102.3°F). Skin rash is absent. Auscultatory findings are unremarkable. All other physical findings are normal. Suspecting acute Q fever, serologic testing by immunofluorescence assay is performed. Results indicate *Coxiella burnetii* infection.

A drug from what class of antibiotics is preferred for the treatment of this zoonotic infection (Q fever) to shorten the duration of fever and reduce the risk of chronic disease (the goals of therapy).

What class and what is the drug?

A. Choramphenicol
B. Cyclic lipopeptides
C. Glycopeptides
D. Lincosamides
E. Macrolides
F. Oxazolidinones
G. Penicillins
H. Streptogramins
I. Tetracyclines*

Hint: Here is an approach to differentiating the correct answer from the distractors:

Recognize the important clues: Recent visit to sheep farm. A zoonotic infection? Confirmed.

Think about the class of drugs used in the treatment of many zoonoses. You don't need to know the specific bug at this time.

Why is knowing the preferred treatment for zoonotic infections important?

Rationale: I. Tetracyclines class.

Doxycycline is first line for the treatment of ***many zoonotic infections***.

For context only. You do not have to memorize Q fever or its cause for this course.

Microbiology: *C. burnettii* is a small Gram-negative coccobacillus obligate intracellular bacterial pathogen. It has a biphasic developmental cycle and is highly resistant to environmental pressures.

Q fever zoonosis: Q fever is transmitted from animals (mainly cattle, sheep, and goats) to humans by aerosol inhalation, rather than arthropod bite. *C. burnetii*'s is an intracellular parasite with a primary growth niche is alveolar macrophages.

Interstitial pneumonia may be present. Myocarditis, pericarditis, and endocarditis are rare, potential complications. The disease is often self-limited. Clinical signs are often very mild or absent. Patients may present with acute self-limited flu-like illness, pneumonia, or hepatitis.

Why is it important to be familiar with treatment for zoonoses?

People travel. People go hiking. People go camping. People are exposed to animals and vectors of zoonotic infections. Always take a thorough history including travel and recreation history.

Incorrect options:

- Chloramphenicol is effective. However, due to potential bone marrow toxicity, it is last-line only for life-threatening infections when other options are not available or not tolerated by the patient.
- Cyclic lipopeptides, like daptomycin, are effective only for Gram-positive infections.
- Glycopeptides, like vancomycin, are effective only for Gram-positive infections.
- Lincosamides, i.e. clindamycin, is useful for susceptible streptococci and staphylococci (Gram-positive aerobes) and some anaerobic Gram-positive and Gram-negative bacteria, including susceptible Bacteroides (Gram-negative). Resistance rates are rising, particularly among Gram-negative organisms.
- Macrolides are for the treatment of Gram-positive infections, atypical pneumonias, nongonococcal sexually transmitted diseases (STIs), and Mycobacterium avium complex pulmonary infection (AIDS patients).
- Oxazolidinones, like linezolid, are effective only for Gram-positive infections.
- Penicillins, all beta-lactams, are ineffective against obligate intracellular infections, such as *C. burnetii* and rickettsial infections.
- Streptogramins, like quinupristin-dalfopristin, are effective only for Gram-positive infections.

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26. A 38-year-old male with extensively drug-resistant (XDR) tuberculosis is given a treatment regimen that includes linezolid. Monitoring includes signs and symptoms of linezolid toxicity that can occur with long-term use of the drug.

What are the potential adverse drug events associated with this drug? (Select all that apply.)

A. Myelosuppression*
B. Optic neuropathy*
C. Peripheral neuropathy*
D. *C. difficile*-associated diarrhea*

Hint: Notes p29.

Rationale: All options are correct. Long-term linezolid use may be limited by toxicities.
The mechanism of toxicity is believed to be inhibition of mitochondrial protein synthesis.

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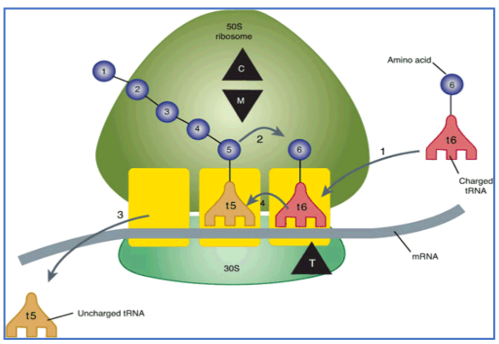
27. What is the mechanism of action of the drug in the above question (#26)?
One sentence or draw and illustration.

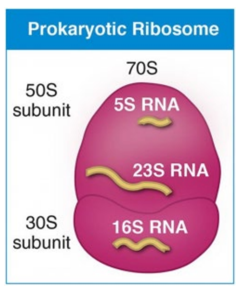
Rationale:

Mechanism: Linezolid binds to a unique site in the A-site pocket on the 23S rRNA of the 50S subunit at the peptidyl transferase center → perturbs the peptidyl transferase center, which affects tRNA positioning → inhibits the binding of the initiator tRNA at the A site (ribosome-fMet-tRNA complex) → prevents the formation of the functional 70S (fMet: N-formylmethionine)

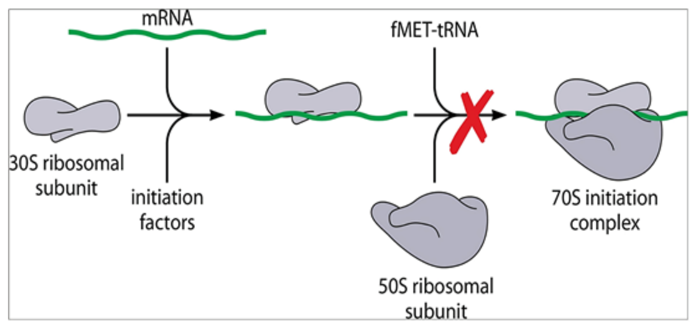
Resistance mechanism: Point mutations in linezolid binding site on 23S ribosome and efflux



Katzung, 13e Figure 44-1
Numbered circles represent amino acids; T: tetracycline binding site; C: chloramphenicol site; M: macrolides site



researchfeatures.com /novel-target-ribosomal-antibiotics/



https://www.scirp.org/journal/paperinformation?paperid=122652

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28. What are the spectrum of activity of the oxazolidinone antibiotics and their place in therapy?

Rationale:

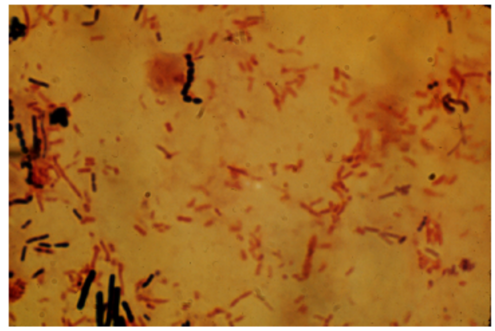
Narrow spectrum of activity: Gram-positive bacteria and *Mycobacterium*

- Gram-positive cocci – staphylococci (MRSA), streptococci, enterococci (*E. faecalis* and *E. faecium*)
- Gram-positive rods – *Listeria monocytogenes, Corynebacterium* spp, *Nocardia* spp
- Gram-positive anaerobes
- *Mycobacterium* spp

Therapeutic uses: "Reserve antibiotic" for complicated and uncomplicated infections by resistant strains: MRSA, VRSA, VRE, *S. pneumoniae*, *Mycobacterium tuberculosis*

It is "reserved " in order to conserve its effectiveness.

Mixed flora in peritoneal fluid from a ruptured viscus



Gram stain of peritoneal fluid (x1000) shows several different organisms, including gram-positive cocci in chains, gram-positive rods, plump enteric gram-negative bacilli, and thinner gram-negative rods. Mixed fecal flora grew from this specimen.

Courtesy of Harriet Provine. UpToDate®

29. An adult patient with a complicated intra-abdominal infection is given treatment with intravenous tigecycline targeting Gram-negative coliforms, streptococci, enterococci, and anaerobic bacteria.

i) What is an advantage of this drug related to drug design over other drugs in the same class?

Short answer.

ii) Which of the following is the most likely drug-specific reason some experts avoid the drug?

A. Concern for increased incidence of mortality associated with use of this drug*
B. Substantial rates of in vitro resistance to *B. fragilis*
C. Frequent development of nephrogenic diabetes mellitus
D. Requirement to avoid the drug in patients with renal impairment.

Rationale:

i) Drug design: Effectiveness against multidrug resistant pathogens

Tigecycline and its derivatives eravacycline and omadacycline were developed for their broad spectrum and activity against MDR strains that express ribosomal protection proteins and many efflux pumps. Note: These drugs are not active against *Pseudomonas*.

ii) A. Concern for increased incidence of mortality

Important caution: Increased all-cause mortality was observed in meta-analyses of phase 3 and 4 clinical trials in tigecycline-treated patients vs. comparator.

The cause of the mortality risk difference has not been established.

Tigecycline and is not indicated for hospital-acquired or ventilator-associated pneumonia.

Incorrect options:

– B. Tigecycline remains, as of now, active against B. fragilis.
– C. Tigecycline is not associated with causing nephrogenic diabetes insipidis.
  Demeclocycline does cause DI.
– D. Tigecycline is metabolized in the liver (phase 2) and excreted in the feces. Very little active drug is excreted in urine. Dosage adjustment is not required for patients with renal impairment.

Tigecycline cautions

Tigecycline and derivatives are associated with numerous adverse effects, including anaphylaxis, hepatotoxicity, pancreatitis, photosensitivity, increased bleeding risk (coagulopathy)

30. A 44-year-old otherwise healthy person presents to the family physician with s/s of rhinosinusitis of 15 days duration. The symptoms initially improved after one week but reoccured 2 days later. The patient reported an allergic reaction (rash) to Pediazole (erythromycin-sulfisoxazole) as a child and a reaction to amoxicillin 6 months ago (urticaria and angioedema). Based on history and physical findings, an e-prescription for levofloxacin is submitted for presumptive acute bacterial rhinosinusits.
After taking the drug as directed for 7 days, the patient complains of pain, burning, tingling, numbness, weakness, and a change in sensation to touch and temperature in both hands and both feet.

What drug-related adverse effect is manifested by these signs and symptoms?

A. Hypersensitivity reaction
B. Hypoglycemia
C. Peripheral neuropathy*
D. Rhabdomyolysis

Students: You should know these terms even though this is your first term of medical school. If you do not know what the terms mean, please do a quick Internet search of the definitions.

AI can help but be sure the information is accurate before relying on it.

Rationale: Peripheral neuropathy (US Boxed Warning)

- Fluoroquinolones are associated with disabling and potentially irreversible neurologic adverse reactions that may occur together. This is a well-described adverse effect, although it is uncommon.

These include:

- Tendinopathy and tendon rupture
- Peripheral neuropathy
- CNS effects: Seizures, increased intracranial pressure, dizziness, lightheadedness, tremors.

Microbiology: *S. pneumoniae* and *H. influenzae* are commensal flora of the upper respiratory tract and the most common causes of acute bacterial rhinosinusitis (33% and 32% of cases, respectively).

- **Practice point:** A respiratory fluoroquinolone is an effective option for penicillin-allergic patients. However, due to the risk of serious FQ side effects, it should be reserved for those who have no alternative treatments.

Respiratory fluoroquinolone definition: FQ active against the most common respiratory pathogens: *S. pneumoniae, H. influenzae, M. catarrhalis*, and the atypical respiratory pathogens.

They are: Levofloxacin, Moxifloxacin, Delafloxacin

Incorrect options:

- A. Hypersensitivity reactions signs/symptoms are usually characterized by an immunological response, such as urticaria, pruritis, angioedema, and/or bronchoconstriction. Allergic reactions can occur with any medication in the susceptible individual.
- B. Hypoglycemia symptoms include dizziness, weakness, palpitations, sweating, and anxiety. Fluoroquinolones have been associated with hypo- and hyperglycemia.

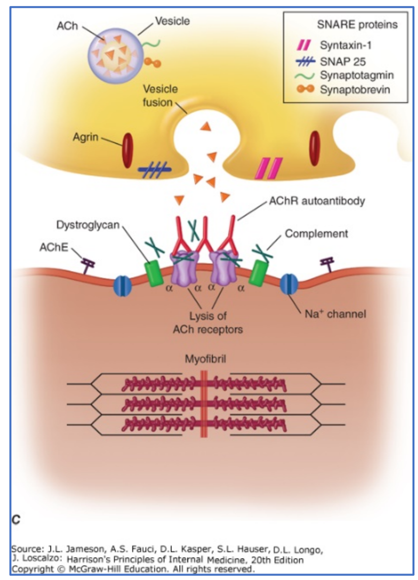
– D. Rhabdomyolysis symptoms include muscle pain, weakness, and dark urine. High doses of statins (drugs for high cholesterol) can cause lysis of skeletal muscle cells and release of cellular constituents into the circulation.

---

31. A 34-year-old woman with a serious Gram-negative infection is evaluated for empiric therapy that includes an aminoglycoside.

This drug should be avoided (if possible) if the patient has:

A. Chronic hepatitis C viral infection (liver disease)
B. Gallstones (potential biliary obstruction)
C. Myasthenia gravis (neuromuscular disorder)*
D. Warfarin therapy (anticoagulant; high potential for DDIs)



Rationale: C. Myasthenia gravis

Drug-disease interaction: Aminoglycosides can cause neuromuscular blockade, which can exacerbate muscle weakness in patients with myasthenia gravis (MG).

Proposed mechanism: Aminoglycosides appear to inhibit the release of the neurotransmitter acetylcholine from the presynaptic nerve terminal in the neuromuscular junction by blocking the voltage-gated calcium channels on the presynaptic membrane.

The function of this VG Ca^{2+} channel is transiently increasing intracellular calcium, which triggers vesicle fusion and release of the neurotransmitter.

Incorrect options

– Aminoglycosides are not metabolized and do not cause hepatotoxicity.
– Aminoglycosides are not excreted via the biliary system.
– Aminoglycosides are not substrates of metabolic enzymes or transporters.
  Drug-drug interactions are related to their toxicities – drugs with similar toxicities increase the risk of adverse drug reactions.

Myasthenia gravis: MG is an autoimmune neuromuscular disorder affecting nicotinic receptors characterized by fluctuating skeletal weakness involving the ocular, bulbar, limb, and respiratory muscles.

Figure448-1c Illustration of a myasthenic neuromuscular junction from Harrison's Principles of Internal Medicine, 21e, 2022; Chapter 448: Myasthenia Gravis and Other Diseases of the Neuromuscular Junction

Legend: The myasthenia gravis (MG) junction demonstrates a reduced number of acetylcholine receptors (AChRs); flattened simplified postsynaptic folds; and a widened synaptic space.

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32. A 22-year-old person presents to the family physician with a low-grade fever, cough, myalgias, and fatigue. Chest x-ray shows diffuse interstitial infiltrates consistent with atypical pneumonia caused by *Mycoplasma pneumoniae*. The patient is prescribed fluids, rest, ibuprofen or acetaminophen, and an oral antibiotic effective against the bacteria that cause atypical pneumonia and that has a relatively low risk of serious toxicities.

What is the drug?

A. Amoxicillin
B. Cefuroxime
C. Chloramphenicol
D. Clarithromycin*
E. Linezolid

Rationale: Macrolides are effective against atypical organisms: *Legionella* spp, *Chlamydia*, and *Mycoplasma.* (*M. pneumoniae* resistance against macrolides is rising.)

Azithromycin or clarithromycin are typically prescribed.

Incorrect options:

- A/B. Amoxicillin / Cefuroxime: Beta-lactams do not accumulate intracellularly. That is, they are not able to pass through the bacterial cell plasma membrane, so they are not effective against obligate intracellular organisms.

  Also, cell wall inhibitors are ineffective against organisms that do not have a peptidoglycan cell well, such as *Mycoplasma* spp.
- C. Chloramphenicol has a broad spectrum of activity. However, it causes does-related suppression of red cell production and is associated with rare but life-threatening aplastic anemia, which limit its use to diseases that have no alternatives.
- E. Linezolid: Active only against Gram-positive bacteria.

---

33. What other classes of drugs would be effective in the treatment of community-acquired pneumonia caused by susceptible atypical bacteria and are generally tolerable?

Make a list.

Rationale: Doxycycline and the respiratory fluoroquinolones (levofloxacin, moxifloxacin, delafloxacin) Minocycline would be effective but is not typically prescribed due to side effect profile.

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34. A 45-year-old patient (pronouns he, him, his) presents to the PCP with complaint of occasional nausea and burning epigastric pain 45 minutes after ingesting a meal. He denies the use of aspirin or other NSAIDs, alcohol, and tobacco or vaping products. Past medical history is significant for a particular congenital problem diagnosed when he was 17 years old.

Presenting signs and symptoms are consistent with a peptic ulcer. The urease breath test is positive for *Helicobacter pylori*. Upper endoscopy confirms the diagnosis of benign peptic ulcer disease.

The patient is evaluated for Clarithromycin + Amoxicillin + Lansoprazole (Prevpak®) for eradication of *Helicobacter pylori*. However, the macrolide in the regimen should be avoided due to the patient's congenital abnormality.

Thinking about the potential serious toxicities of the macrolide in the regimen and the patient's hereditary disease, **what is the most likely reason for avoiding this peptic ulcer therapeutic regimen in this patient?**

A. Agammaglobulinemia (immune deficiency)
B. G6PD deficiency (metabolic disorder)
C. Long-QT syndrome (electrophysiological disorder)*
D. Tourette syndrome (neurological disorder)

Hint: Notes p17.

Rationale: C. Long QT syndrome

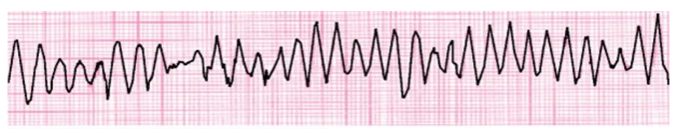
Combining drugs that cause QT prolongation increases the risk of *life-threatening arrhythmias*.

I have provided different scenarios in which patients might develop TdP so you can avoid it in your future practice (and get the Board questions right).

This question describes the QT prolongation adverse effect in the context of a congenital cardiac abnormality caused by mutations in genes that encode cardiac ion channels or their regulatory proteins.

- **Drug-induced prolongation:** Congenital long QT syndrome may ***manifest for the first time*** when a patient with the underlying genetic predisposition is exposed to a drug that prolongs the QT interval.

Drugs that cause QT interval prolongation in susceptible patients can induce a life-threatening ventricular arrhythmia called torsades de pointes – twisting of the points, as seen on EKG:



Source: Knoop KJ, Stack LB, Storrow AB, Thurman RJ: *The Atlas of Emergency Medicine, 3rd Edition:* http://www.accessmedicine.com

The macrolides are associated with causing QT interval prolongation. This adverse drug effect occurs rarely in patients who do not have risk factors.

- **Practice points:** Risk factors for QT interval prolongation and torsades de pointes include congenital and acquired long QT syndrome.

Risk factors for acquired LQTS include:

- Drugs: Macrolides, azole antifungals, antiarrhythmics, some antipsychotics, some antidepressants, and others – especially high doses, rapid I.V. infusions, or concomitant use
- Enzyme inhibitors: Other drugs that inhibit the metabolism, and increase systemic levels, of the QT interval prolongers
- Electrolyte disturbances, primarily hypokalemia and hypomagnesemia
- Underlying structural heart disease
- Advanced age

Incorrect options: Macrolides are not associated with exacerbating the disorders listed in the other options.

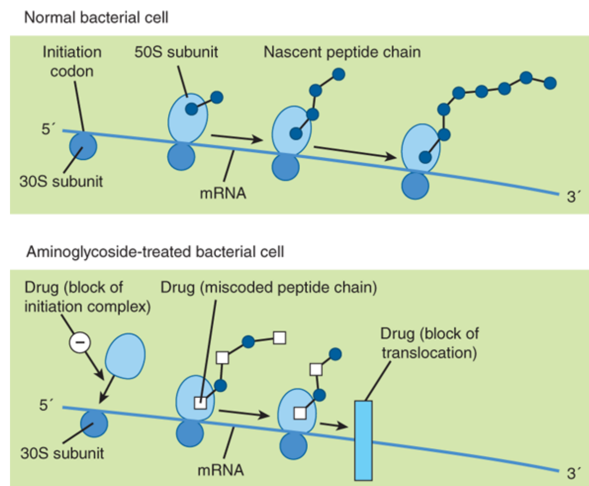
EKG: The Atlas of Emergency Medicine > Part 2. Specialty Areas > Chapter 23. ECG Abnormalities > Part 3: Rhythm Disturbances > Torsades De Pointes > ECG Findings

---

35. Which of the following options best represents the mechanism of action of the aminoglycosides?

A. Blocking the initiation complex
B. Eliciting premature protein synthesis termination
C. Inducing accumulation of nonfunctional monosomes
D. Misreading mRNA
E. Incorporating incorrect amino acids
F. All of the above*

What are the aminoglycosides' binding site and its physiologic function?



Source: Todd W. Vanderah: Basic & Clinical Pharmacology, Sixteenth Edition

Aminoglycosides & Spectinomycin, Vanderah TW. Katzung's Basic & Clinical Pharmacology, 16th Edition; 2024

Rationale: Aminoglycosides are irreversible inhibitors of bacterial protein synthesis.

The actions were identified in studies on streptomycin. It is believed that the other AGs have similar actions.

The actions occur more or less simultaneously, irreversibly, and lead to cell death.

Aminoglycosides bind to the 16S rRNA on the 30S subunit. The 16S rRNA is required to initiate protein synthesis and stabilize correct codon-anticodon pairing at the A site during mRNA translation

Aminoglycosides bind to polysomes and interfere with protein synthesis by the following mechanisms at least (there may be more):

a) Interfering with the initiation complex of protein synthesis,
b) Misreading and incorporation of incorrect amino acids resulting in production of abnormal nonfunctional proteins,
c) Blocking translocation on the mRNA

Other suggested mechanisms:

The aberrant proteins may be inserted into the cytoplasmic membrane causing leakage of small ions, followed by larger molecules, leading to cell death.

Breakup of polysomes into nonfunctional monosomes may occur by blocking movement of the ribosome after the formation of a single initiation complex, resulting in and mRNA chain with a single ribosome on it.

---

36. A 72-year-old woman with severe chronic obstructive pulmonary disease (COPD) presents to her pulmonologist for evaluation of frequent exacerbations, despite optimal management. After weighing the potential benefits against the risks of long-term macrolide use, prophylactic macrolide antibiotic

therapy is considered for this patient. The patient's medication history is significant for the anticoagulant apixaban for atrial fibrillation. Erythromycin and clarithromycin are contraindicated due to a metabolism and transport related drug-drug interaction with the apixaban.

i) What are the metabolic- and transport-mediated drug interactions that could lead to potentially severe bleeding (an effect of excess systemic levels of the anticoagulant)?

Highlight the two correct answers.

A. CYP1A2	a. OCT
B. CYP2D6	b. GLUT4
C. CYP3A4	c. P-gp
D. UGT1A1	

ii) What is a potentially serious adverse drug event that could occur as a result of this drug-drug interaction?

Hint: Pick out what you *need* to know and what you *do not need* to know.

The information in the stem indicates that there is a potentially serious drug interaction with the apixaban.

To answer this question, you do not need to know the pharmacokinetic properties of the apixaban because you can predict and deduce the potential adverse drug event of the interaction in this case by knowing the potential adverse effects of the apixaban and the metabolism- and transport-related interactions caused by the macrolide antibiotic.

Of course, in practice, you will always look it up in the drug monographs to be sure.

Rationale:

i) Erythromycin and clarithromycin are substrates and strong inhibitors of CYP3A4. As such, there is a high potential for drug-interactions involving them. (Drug Interactions lecture).

Inhibiting CYP3A4 increases plasma levels of the substrate drug.

- Since it is stated in the stem that apixaban should not be given concomitantly with erythromycin or clarithromycin due to a metabolism/transport-related drug-drug interaction, apixaban must therefore be a CYP3A4 substrate.

ii) Inhibiting the metabolism of apixaban can lead to elevated systemic concentrations (increased bioavailability) of the apixaban, which could result in excessive bleeding. If hemorrhage occurs in the brain, the patient could suffer a stroke.

- **Practice point:** The goal of prophylactic antimicrobial therapy is to prevent infection using an antimicrobial targeted against the potential pathogens.

Strong data implicates bacterial infection as a precipitant of a substantial portion of exacerbations in patients with COPD (Harrison's).

37. After checking the Drug Interactions Tool in UpToDate, an e-prescription is submitted for a macrolide that does not inhibit CYPs or inhibit renal transporters.

i) What antibiotic is selected for the COPD patient in the above question? Short answer

ii) What are the clinically relevant pharmacokinetic properties of this drug?

Azithromycin-specific takeaways: Reminder

- Once daily dosing, sequestered in tissues
- Dosing for renal insufficiency not necessary
- Long half-life allows prolonged antimicrobial effect, even after the patient is finished taking the medication.
- Azithromycin is a weak inhibitor of CYP3A4.
- Cautious use in patients with hepatic impairment
- Azithromycin DOES cause QT interval prolongation.

Rationale:

i) Azithromycin is a weak CYP3A4 inhibitor, meaning it does not usually cause major drug interactions and does not usually require dose adjustments for CYP3A4 substrates.

FYI: The drug developer of apixaban, Eliquis, found no pharmacokinetics interaction with azithromycin.

(Eliquis package insert, revised Nov 2019; Bristol-Myers Squibb)

ii) PK properties:

Oral – useful for home administration (or IV – useful for hospital administration)

Widely distributed, concentrates in lung tissues and phagocytes,

- High intracellular concentrations → extensive tissue sequestration → long half-life as the drug is slowly released from sequestered sites → sustained antimicrobial activity after drug is discontinued

Excreted through the biliary system (~50% unchanged drug) and undergoes some hepatic metabolism.

Adequate bioavailability

〉You may be wondering what are the risk associated with long-term use of azithromycin for prevention of COPD exacerbation. They include development of resistance, QT interval prolongation and arrhythmia, C. difficile infection, cholestatic hepatitis, hearing loss. Adverse effects may be mitigated by monitoring for them.

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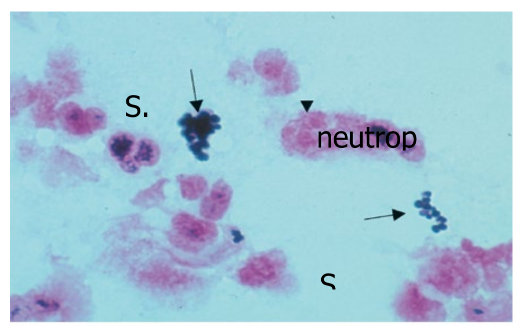
38. Janie is 20-year-old person who presents to the PCP with complaint of a "puss-filled sore" on their left elbow. It began as an abrasion that has gradually gotten worse over the past week. The patient has not had a fever or any other systemic symptoms.



Source: Kenneth J. Ryan: Sherris Medical Microbiology, Seventh Edition

The abscess is debrided and a sample of exudate is sent for culture.

(Debride pronunciation: de breed')

Gram stain shows Gram-positive cocci in clusters resembling bunches of grapes (arrows) and neutrophils (arrowhead).

An e-prescription is submitted for empiric therapy with an oral protein synthesis inhibitor. The drug:

- Accumulates in phagocytes and wound exudates
- Inhibits peptidyl transferase activity, which prevents bond formation with the incoming amino acid,
- Has good activity against community-acquired MRSA (caMRSA).

What is the drug?

A. Ciprofloxacin
B. Clindamycin*
C. Erythromycin
D. Sulfadiazine
E. Tetracycline

Rationale: B. Clindamycin fits the description.

Clindamycin is effective against uncomplicated skin/soft tissue infections caused by susceptible MRSA (negative D-test).

Incorrect options:

- A. Ciprofloxacin: DNA gyrase inhibitor; not effective against MRSA
- C. Erythromycin: Slightly different mechanism of action and not effective against MRSA
  Macrolides bind the 23S near the peptidyl transferase center (like clindamycin) and block the exit tunnel, which prevents the forming polypeptide from leaving the ribosome. Protein synthesis comes to a halt.
- (Silver) Sulfadiazine: A sulfonamide – folate synthesis inhibitor. Silver sulfadiazine is a topical cream for external use. It has activity against MRSA but is used for burn wound care.
  NOTE: TMP-SMX is effective against MRSA and is a first-line oral option for skin/soft tissue MRSA infections.
- Tetracycline: Binds 16S rRNA on 30S subunit and blocks the binding site (A site) of incoming tRNA.
  NOTE: Doxycycline and minocycline are effective against MRSA and are first-line oral options for skin/soft tissue MRSA infections.

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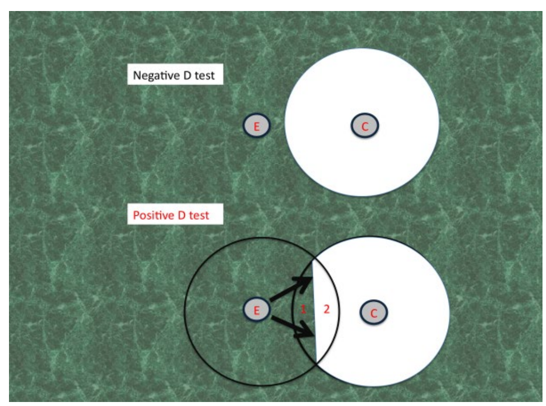
39. Janie initially responds to antibiotic therapy. D-test result obtained the following day is positive, which suggests that a subpopulation of MRSA resistant to Janie's prescribed antibiotic may lead to clinical failure or recurrence of the infection after therapy is discontinued.



i) What are the gene family and the expressed protein?

ii) What bacteria may harbor these genes?

iii) What is the mechanism of resistance?

iv) What three classes of drugs are cross resistant?

Rationale:

i) The stem describes modification of the ribosomal binding site by methylase expressed by the *erm* genes *erm*A, *erm*B, *erm*C . erm= erythromycin ribosomal methylase

ii) *Staphylococci, Streptococci*, and *Enterococci* (and certain others) may harbor *erm* genes.

iii) Acquisition of erm genes, by plasmid transfer or integrated into bacterial chromosomal DNA and transfer to daughter cells.

Production of methylase can be inducible or constitutive.

- Inducible *erm*: expression of erm gene in the presence of an inducer, such as a macrolide
- Constitutive *erm*: Continuous expression of ribosomal methylase even without antibiotic exposure

❖ **Mechanism:** Macrolides induce expression of the erm resistance genes, which produces the methylase, which methylates the bacterial ribosome's 23S rRNA, preventing binding of the drugs.

Clindamycin is not an inducer of the methylase gene.

However, clindamycin treatment of a Gram-positive infection – notably *S. aureus*, either MSSA or MRSA, or *S. pneumoniae* – harboring the $iMLS_B$ may cause resistance to emerge *during therapy*.

Failure to eradicate the infection or disease recurrence may result.

➢ **Practice point:** Strains harboring $iMLS_B$ *appear susceptible* to clindamycin in vitro but resistance may emerge during therapy.

A positive D test suggests that a subpopulation of microbes resistant to clindamycin may emerge and lead to clinical failure or recrudescence and should be reported as clindamycin resistant.

The Clinical and Laboratory Standards Institute recommends testing for inducible clindamycin resistance ($iMLS_B$) on all *Staphylococcus* spp and *Streptococcus pneumoniae* spp.

iv) MLSB phenotype: Cross-resistance occurs to all macrolides, clindamycin (lincomycin class) and streptogramin B (quinupristin)

D test: Figure: TUSOM Pharmwiki http://tmedweb.tulane.edu/pharmwiki/doku.php/the_d_test

Legend: Figure Legend.

Top Panel: Shows a stylized cartoon of a negative D-test observed for an erythromycin-resistant culture of *S. aureus*. The small discs labeled E & C represent disks containing either 15 µg erythromycin (E) or 2 µg clindamycin (C) placed 15 to 20 mm apart on an agar plate that has been inoculated with the clinical isolate (indicated by the green background). The lack of a zone of inhibition around the erythromycin disc indicates bacterial resistance to macrolides (e.g. perhaps due to expression of a P-glycoprotein efflux pump that affects macrolides). The large clear zone of inhibition around the clindamycin disc indicates sensitivity to clindamycin.

Bottom Panel: Depicts a positive D-test. Diffusion of erythromycin from the disc towards the clindamycin disc does not kill bacteria due to *S. aureus* resistance to macrolides. However, in this case the bacterial isolate contains a strain of *S. aureus* with an erythromycin-inducible methylase (iMLS-B) that is encoded by a plasmid-borne gene (erm). When this methylase is induced it alters the binding site on the 23S subunit of the 50S ribosome that both erythromycin and clindamycin bind to, making both antibiotics ineffective (inducing resistance). As a result, as erythromycin diffuses outward into Zone 1 bacterial resistance to clindamycin is induced prior to its diffusion from the neighboring disk. In contrast, growth inhibiting concentrations of clindamycin reach zone 2 before erythromycin can arrive to induce resistance (due to the shorter distance for diffusion), resulting in inhibited growth.

The inhibition of bacterial growth in zone 2 but not zone 1 produces a “D” shape surrounding the clindamycin disk, which is considered a “positive” D-test. (Adapted from Woods 2009).

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40. A 28-year-old patient in the burn unit is treated for severe thermal burn injury and sepsis. NKDA. Management includes empiric broad-spectrum antibiotic therapy and debridement of the necrotic tissue. Based on the hospital’s antibiogram, a 3-drug regimen is selected for coverage of presumptive MRSA and other Gram-positive bacteria, enteric aerobic Gram-negative bacilli (GNBs), and multidrug-resistant *P. aeruginosa*.

What antibiotic therapy is being given to this patient?

Highlight the preferred agent from each column (one choice each).

A. Aerobic Gram-negative bacilli and *P. aeruginosa*	B. Broad-spectrum agent with activity against aerobic GNBs and *P. aeruginosa*	C. Anti-MRSA
Amikacin Gentamicin Neomycin Paromomycin Tobramycin	Ampicillin-sulbactam Ceftriaxone Clindamycin Meropenem Tigecycline	Cefazolin Daptomycin Linezolid Quinupristin-Dalfopristin Vancomycin

Rationale:

A. Aerobic Gram-negative bacilli and *P. aeruginosa*	B. Broad-spectrum agent with activity against aerobic GNBs and *P. aeruginosa*	C. Anti-MRSA
Amikacin Gentamicin Neomycin Paromomycin Tobramycin	Ampicillin-sulbactam Ceftriaxone Clindamycin Meropenem Tigecycline	Cefazolin Daptomycin Linezolid Quinupristin-Dalfopristin Vancomycin

A. Tobramycin has somewhat greater intrinsic activity against *P. aeruginosa* than gentamicin but gentamicin is also effective.

- Like gentamicin, tobramycin is effective against aerobic GNB. Amikacin is reserved for strains resistant to tobramycin or gentamicin.

Students, If you selected gentamicin or amikacin, you may consider yourself to be correct.

- **Practice point:** You should strive to recognize the nuances between drugs because being able to do so will help you choose which drugs to prescribe for which therapeutic goals.

B. Meropenem is a carbapenem with broad-spectrum activity against Gram-positive (not MRSA) and Gram-negative bacteria, including

- *P. aeruginosa* is intrinsically resistant to the other agents in column B.
- Ampicillin-sulbactam and ceftriaxone are "extended-spectrum" antibiotics, meaning they are more active against Enterobacterales species than there in-class predecessors.
- Clindamycin is used in the treatment of susceptible Gram-positive infections and anaerobic infections.

C. Vancomycin is the drug of choice for the intravenous treatment of susceptible MRSA.

- Daptomycin and linezolid are effective against MRSA and are alternatives when vancomycin cannot be used due to resistance or other reason.
- Quinupristin-dalfopristin are streptogramins in fixed-combination for synergy and bactericidal action against aerobic Gram-positive bacteria only. Marketing has been discontinued but the drug could still appear on a board exam.

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41. A 26-year-old patient presents to the ED with fever, cough, and shortness of breath. Chest radiograph shows diffuse bilaterally symmetric interstitial pattern opacification and multiple small hyperlucent patches. HIV screening is positive. The patient is referred to the infectious disease specialist and ultimately given a diagnosis of HIV/AIDS (CD4 cell count 150 cells/mm^3) and *Pneumocystis* pneumonia, a fungal infection.

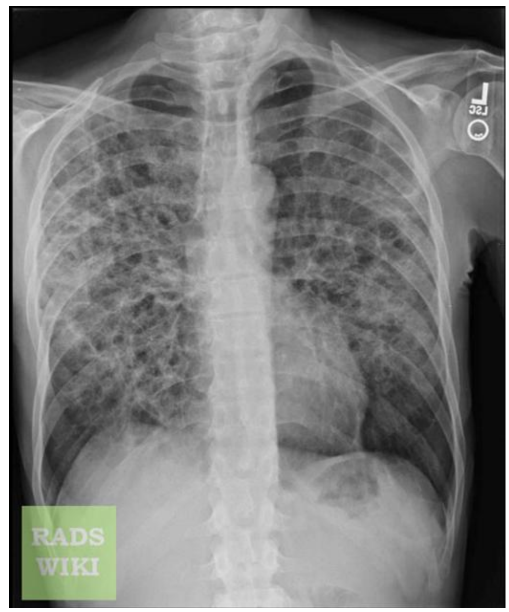
Antiretroviral therapy and the drug of choice for *Pneumocystis jirovecii* pneumonia treatment (and prophylaxis) are initiated.

The patient presents 7 days later with a rash that appeared suddenly this morning. Suspecting an exanthematous drug eruption, the patient is directed to stop taking the drug for the pneumonia treatment. The rash resolves gradually over the course of the next 5 days. He is placed an alternative antimicrobial regimen.

What drug is the preferred treatment for *Pneumocystis* pneumonia that likely caused the rash?

A. Amphotericin B
B. Ciprofloxacin
C. Doxycycline
D. Itraconazole
E. Trimethoprim-Sulfamethoxazole*

Hint: Handout p5-6



Diffuse bilaterally symmetric interstitial pattern is noted in the perihilar region and extending towards the periphery. Multiple ill-defined small hyperlucent patches are noted in the bilateral lung fields especially in the mid-zones suggestive of pneumatocele.

Numerous small, discrete papules on the back of a patient with exanthematous drug eruption.

Leslie Goldstein, Pharm.D.
Updated September 2025

Rationale: TMP-SMX is the most effective regimen for the treatment and prevention of *Pneumocystis jirovecii*, which is a fungal infection.

- The frequency of hypersensitivity reactions in patients with HIV/AIDS is 30%-50% compared to non-HIV positive controls (3%). The mechanism is unclear.

  Susceptibility to oxidative damage due to a general state of glutathione deficiency and slow acetylator phenotype have been hypothesized but not confirmed empirically.

Upper figure: Radswiki T, Pneumocystis carinii pneumonia. Case study, Radiopaedia.org (Accessed on 13 Aug 2024) https://doi.org/10.53347/rID-11789

Lower figure: UpToDate Graphic 86552 Version 8.0: Exanthematous (morbilliform) drug eruption

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42. In addition to skin rashes, the hypersensitivity reaction to the sulfonamide may progress. What other manifestations may occur? (More than one answer. Choose all that apply.)

    A. Kidney: Crystalluria
    B. Liver: Hepatitis*
    C. CNS: Kernicterus
    D. Skin: Stevens-Johnson syndrome*
    E. Bone marrow: Thrombocytopenia*

Definition: Morbilliform (looks like measles) exanthem (rash); macular lesions.

Rationale: Skin rash with fever may progress to multisystem organ involvement:

Stevens-Johnson syndrome/toxic epidermal necrolysis, hepatitis, nephritis (inflammation due to immune attack, not due to crystalluria), pulmonary infiltrates, and cytopenias

These effects can range from mild to severe to fatal.

Incorrect options:

- Crystal formation in the urinary tract due to sulfonamide precipitation in neutral or acid pH with sulfonamides, including sulfamethoxazole. This effect occurs mainly with the insoluble sulfonamides and is related to the sulfonamide chemical properties.
- Patients taking sulfonamides should be advised to drink plenty of fluids.
- Kernicterus is an encephalopathy – bilirubin-induced neurological dysfunction – in newborns. Premature neonates, who have immature glucuronosyl transferase capacity and blood brain barrier, are more vulnerable than term infants and at lower total serum bilirubin levels. It is caused by deposition of unconjugated bilirubin in several areas of the brain, causing neurological damage.

Mechanism of kernicterus adverse drug effect: Sulfonamides displace unconjugated bilirubin from albumin binding sites, increasing free concentrations in plasma.

43. What is an important metabolic enzyme primarily involved in the hepatic biotransformation of sulfonamides to inactive metabolite that is subject to genetic polymorphisms resulting in slow, intermediate, and rapid metabolizer phenotypes?

Short answer

Rationale: N-acetyltransferases

Phase 2 reaction: N-acetyltransferases (NAT) metabolize aromatic amines by transferring an acetyl group from acetyl-CoA to the nitrogen, N, moiety on the drug molecule (center structure in the figure) $\rightarrow$ nontoxic N4-acetyl sulfonamide.

CYP2C9 pathway contributes to sulfonamide metabolism and generation of toxic N4-hydroxy and nitroso metabolites.

These reactive metabolites may covalently link to large host proteins, causing direct cytotoxicity, or act as haptens and initiate an immune attack .

- Pharmacogenomics: Slow acetylator phenotype appears to be a risk factor for sulfonamide hypersensitivity reactions.

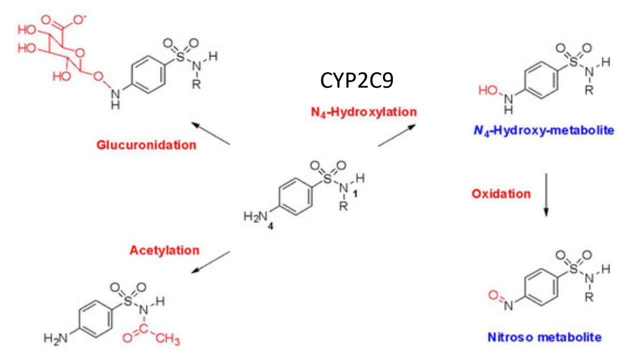

Figure: Giles A, Foushee J, Lantz E, Gumina G. Sulfonamide allergies. Pharmacy 2019, *7*(3), 132; https://doi.org/10.3390/pharmacy7030132 Open Access

44. A penicillin- and cephalosporin-allergic health care worker is treated for a subcutaneous *S. aureus* infection with clindamycin 450 mg po TID x10d. The infection resolves and the patient completes the clindamycin therapy. Two weeks later, the patient calls his physician with complaint of low-grade fever, abdominal cramping, loss of appetite, and diarrhea consisting of 6-8 watery stools per day for 3 days. He denies blood in the stool. Stool cultures reveal an organism known to cause antibiotic-associated diarrhea. Nucleic acid amplification test reveals bacterial toxin.

Definition: po TID: by mouth 3 times daily

What is the diarrhea-causing pathogen in this case?

A. *Clostridioides difficile**
B. *Bacteroides fragilis*
C. *Enterococcus faecium*
D. *Escherichia coli*
E. *Salmonella enterica*

Rationale: Clindamycin has long been associated with causing *C. difficile* overgrowth and subsequent diarrhea and pseudomembranous colitis (patches of abnormal tissue – false membranes – on the lining of the colon.

- **Practice point:** *Clostridioides difficile* is a toxin-producing, spore-forming Gram-positive, anaerobic bacterium that colonizes the intestinal tract after normal gut flora have been disrupted.

C. difficile infection is one of the most common health care-associated infections. It is a significant cause of morbidity and mortality, especially among older adult hospitalized patients. Clindamycin is a frequently implicated cause because of disruption of normal gut flora, particularly anaerobes.

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45. A generally healthy 42-year-old patient, who has been given a clinical diagnosis of acute cystitis, is evaluated for antibiotic therapy effective against *E. coli*. The patient has never before had a urinary tract infection. Medical history is significant for warfarin prophylaxis of venous thromboembolism following knee arthroplasty two weeks ago. The patient does not smoke or use nicotine products, drink alcoholic beverages, or use recreational drugs. NKDA. Trimethoprim-sulfamethoxazole should be avoided in this patient.

What is the reason for selecting an alternative antibiotic instead of TMP-SMX?

A. TMP-SMX could be ineffective in treating urinary tract infections caused by *E. coli*.
B. TMP-SMX could crystallize in renal tubules and cause obstruction.
C. TMP-SMX could increase the risk of excessive bleeding.*
D. TMP-SMX could cause emergence of *S. aureus* resistant strains and cause surgical site infection.

Hint: Notes p11

Rationale: C. bleeding

Sulfamethoxazole can increase the risk of bleeding when taken concurrently with warfarin.

Please see the explanation in the next question.

Incorrect options: The interaction is due to the sulfonamide component.

- A. TMP-SMX is highly effective against susceptible *E. coli* UTIs. The patient is generally healthy and has not previously been treated for an acute urinary tract infection so the pathogen is likely to be susceptible in this patient.
- B. Sulfamethoxazole is less likely to precipitate in renal tubules in acidic urine than the older sulfonamides. That said, patients should be encouraged to drink plenty of fluids while taking the drug.
- D. Emergence of resistant *S. aureus* is documented with TMP-SMX but is not the reason for avoiding the drug in this patient.

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46. What are the postulated mechanisms of the TMP-SMX and warfarin drug-drug interaction in the above scenario?

State 3 possible mechanisms may occur simultaneously.

Hint: Handout p11

Rationale: high risk of increasing warfarin levels and bleeding

1) disturbance of gut microbiota, 2) CYP2C9 inhibition, 3) warfarin displacement from plasma proteins

Sulfonamide-warfarin interaction is likely due to a triple effect:

<table>
<tr><td>killing of gut microbiota
↓
enhanced warfarin action</td><td>+</td><td>CYP2C9 inhibition of warfarin metabolism
↓
increased warfarin levels</td><td>+</td><td>displacement of warfarin from serum albumin by sulfonamide
↓
increased free warfarin</td></tr>
<tr><td colspan="5">Increased risk of excessive bleeding
Monitor clotting time closely
Consider therapy change</td></tr>
</table>

1) Disturbance of gut microbiota:

Warfarin is a vitamin K antagonist. It blocks the synthesis of hepatic coagulation factors by blocking a critical enzyme, vitamin K reductase, which prolongs the time required for a clot to form.

Killing the microorganisms that synthesize vitamin K may enhance anticoagulant effects of warfarin could lead to excessive bleeding.

How?

- We obtain vitamin K in our diet and certain organisms in the gut microbiome synthesize it.
- Vitamin K is required for the synthesis of hepatic coagulation factors necessary for blood clotting.
- Killing gut bacteria can reduces vitamin K levels, which decreases the synthesis of vitamin K-dependent hepatic coagulation factors, which may increase the risk of bleeding.

Mechanism of ***any*** antibiotic and warfarin drug interaction

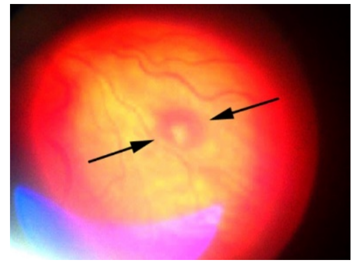
2) CYP2C9 inhibition: Sulfamethoxazole is a substrate and inhibitor of CYP2C9. Warfarin is inactivated by CYP2C9.

3) Sulfonamides displace warfarin from serum albumin binding sites:

- *Warfarin is highly plasma protein bound*, ~99%, meaning about 1% in the blood is unbound = free.
- *Warfarin has a narrow therapeutic window*. Even a relatively small shift from the plasma proteins to unbound drug can increase the risk of bleeding.
  For example, shifting only 1% of warfarin off binding sites increases the free fraction from 1% to 2%, which *doubles* the concentration of drug available to interact with its target.
- Reduced vitamin K levels along with warfarin-inhibition of vitamin K activity further decreases the synthesis of vitamin K-dependent hepatic coagulation factors, which may further increase the risk of bleeding.

✸These are very important mechanisms to remember and understand, even though you haven't had a lecture on warfarin pharmacology.

47. A 45-year-old woman presented to the ED with fever, shortness of breath, persistent back pain, and a 10-pound weight loss. She reported chills, sweats, and anorexia for 6 weeks, which had been progressively increasing in severity, and persistent back pain for 4 weeks. Past medical history was significant for rheumatic fever, characterized by swollen joints and fever, which confined her to bed for 3 months when she was a young child. NKDA.



- VS: T 100.4°F (38°C), pulse 90/min, resp 18/min, BP 130/80 mmHg;
- Roth spot (a round white spot surrounded by hemorrhage) in the retina of her right eye and petechiae were seen in the conjunctiva of both eyes.
- Splinter hemorrhages were seen under two fingernails of her right hand and one finger of the left hand.
- Osler nodes (tender, small, raised, red or purple lesions of the skin) were seen in the pads of one finger and one toe.
- Her heart size was normal to percussion.
- On auscultation, a low-pitched diastolic murmur consistent with mitral valve stenosis was heard at the apex; a loud mitral valve opening snap was heard over the left chest.

  https://www.google.com/search?q=mitral+valve+stenosis+spund&source=lmns&tbm=vid&bih=783&biw=1600&rlz=1C1CHBF_enUS1029US1029&hl=en&sa=X&ved=2ahUKEwiqw_av0rmAAxWzUDUKHa63CpAQ0pQJKAF6BAgBEAQ#fpstate=ive&vld=cid:ae221fbf,vid:5oCPtZo4pUY

Subungual splinter hemorrhages

Subungual splinter hemorrhages in a patient with *Trichinella papuae*.

Courtesy of Peter F Weller, MD.

UpToDate®

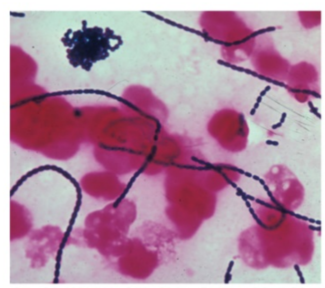
Osl

Osler nodes are tender papulopustules located on the pulp of the finger in a patient with bacterial endocarditis caused by *Staphylococcus aureus*.

Courtesy of Charles V Sanders. The Skin and Infection: A Color Atlas and Text, Sanders CV, Nesbitt LT Jr (Eds), Williams & Wilkins, Baltimore 1995.

http://www.lww.com UpToDate®

- Echocardiography showed an enlarged left atrium, thickened mitral valve leaflets, and a vegetation on the posterior leaflet.
- 3 blood samples were obtained for culture. The patient was admitted for treatment.
- One day later, all three were positive for Gram-positive cocci in chains that were viridans group streptococci subsequently identified as *Streptococcus sanguis*.
- Sensitivity results show that the organism was sensitive to the antibiotics tested but the MIC for the beta-lactam was elevated indicating intermediate resistance.



Source: S. Riedel, J.A. Hobden, S. Miller, S.A. Morse, T.A. Mietzner, B. Detrick, T.G. Mitchell, J.A. Sakanari, P. Hotez, R. Mejia Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e

Streptococci grown in blood culture showing Gram-positive cocci in chains. Original magnification × 1000.

Citation: Chapter 14 The Streptococci, Enterococci, and Related Genera. Riedel S, Hobden JA, Miller S, Morse SA, Mietzner TA, Detrick B, Mitchell TG, Sakanari JA, Hotez P, Mejia R. Jawetz, *Melnick, & Adelberg's Medical Microbiology*, 28e; 2019. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2629&sectionid=217770783 Accessed: November 05, 2020

Antibiotic treatment: 2-drug intravenous antibiotic therapy was initiated.

- One drug was a cell wall inhibitor with a relatively narrow spectrum that is highly active against viridans group streptococci – in fact, highly active against all susceptible streptococci.
- The second drug was able to reach its intracellular site of action because of the breakdown of the cell wall and was added for synergistic effect.

What antibiotics were administered? Highlight the selected agents:

Cell wall inhibitor	Drug for synergy
Aztreonam	Azithromcyin
Cefiderocol	Doxycycline
Imipenem-cilastatin	Gentamicin
Ampicillin	Vancomycin

Hint: Do not be distracted by the name of the specific bacterial pathogen. Focus on the relevant information:

Rationale: Ampicillin plus Gentamicin

Drug selection:

- As a general statement, antibiotic therapy should be a narrow-spectrum *targeted* to the organism isolated from the blood samples.
- Ampicillin is highly active against susceptible streptococci,

  has good penetration into vegetations on heart valves,

  covers a range of streptococcal species including strains that may show reduced susceptibility to penicillin G,

  has a synergistic effect with gentamicin especially against viridans streptococci (and enterococci),

  and is supported by guidelines.
- Gentamicin is the selected agent due to efficacy and long experience with the drug for this indication.

Infective endocarditis is an intravascular infection with the potential for multi-organ involvement.

Incorrect options:

– Aztreonam is active only against aerobic Gram-negative organisms.

– Cefiderocol is active only against aerobic Gram-negative organisms.

– Imipenem-cilastatin is active against viridans strep, but it has a broad spectrum of activity.

  Directed therapy with a narrow-spectrum drug minimizes the development of resistance – a goal of antibiotic stewardship practices.

– Azithromycin and doxycycline have relatively poor activity against Gram-positive bacteria due to resistance.

– Vancomycin is active against Gram-positive organisms and exhibits synergy with gentamicin. It is an alternative in patients with penicillin allergy.

Resources:

Case: Access Medicine Jawetz, Melnick & Adelson's Medical Microbiology, 26e Chapter 48. Cases and Clinical Correlations

Figures

UpToDate graphics Splinter hemorrhages 52141 v. 4.0 and Osler nodes 68796 v.9.0

14-1 Legend: Streptococci grown in blood culture showing Gram-positive cocci in chains. Original magnification × 1000.

Access Medicine Katzung & Vanderah's Basic & Clinical Pharmacology, 15e, 2021; Chapter 45: Aminoglycosides & Spectinomycin

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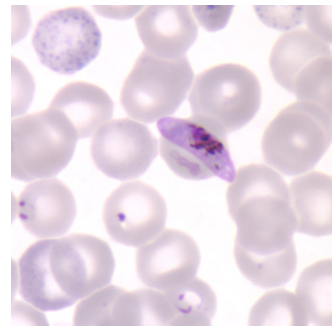
48. A 6-year-old child is brought to the ED with complaints of fever, chills, muscle aches, night sweats, and malaise. His family recently returned to the US from a malaria endemic region. Giemsa-stained blood smears by light microscopy show *Plasmodium falciparum* (malaria). He is given a 7-day course of therapy with quinine and an antibiotic effective in the treatment of malaria.



What is the prescribed drug?

A. Chloramphenicol
B. Clarithromcyin
C. Clindamycin*
D. Doxycycline*
E. Linezolid

Rationale: Clindamycin and doxycycline are both effective in combination with quinine. Doxycycline is preferred.

Traditionally, clindamycin has been recommended for children <8 years old due to tetracycline's recognized staining of the unerupted teeth. Short courses of doxycycline (≤21 days) are now recommended for children <8 years because of doxycycline's low propensity for staining the teeth.

Incorrect options: Clarithromycin, linezolid, and chloramphenicol are not active against *Plasmodium* spp.

CDC DPDx https://www.cdc.gov/dpdx/malaria/index.html Figure D: Gametocyte of *P. falciparum* in a thin blood smear. Also seen in this image are ring-form trophozoites and an RBC exhibiting basophilic stippling (upper left).

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49. A 78-year-old woman presents with severe flank pain, tachycardia, diaphoresis, and nausea. She takes a urinary antiseptic that is hydrolyzed to formaldehyde for prophylaxis of recurrent urinary tract infections. She explained that she is also self-treating acute urinary symptoms that started 3 days ago with TMP-SMX, which she had left over in her medicine cabinet.
Physical examination and urinalysis findings are consistent with crystalluria.

Photomicrograph showing urine sediment of a patient with sulfonamide crystalluria

Urine sediment showing sulfonamide crystals with a needle-shaped appearance. Other forms that may be seen include rosettes and a shock of wheat appearance.

Courtesy of Harvard Medical School. UpToDate®

i) **Which one of the following is the urinary antiseptic?**

A. Ampicillin

B. Cephalexin

C. Ciprofloxacin

D. Methenamine*

E. Sulfasalazine

ii) **What is the mechanism of this drug-induced adverse event?**

Rationale:

Methenamine in acidic urine

↓ Hydrolysis

Formaldehyde is formed and concentrates in renal tubules

(formaldehyde has nonspecific bactericidal action)

↓ Sulfonamide

Sulfonamide + formaldehyde forms an insoluble precipitate in the urine

↓

Crystalluria

↓

may cause urinary tract irritation,

such as tubular injury or obstruction

Methenamine – sulfonamide combination is *contraindicated.*

Incorrect options: The drugs in the other options are antibiotics used to treat various acute infections. FYI: TMP-SMX and cephalexin may be used for UTI prophylaxis.

Figure: UpToDate® Graphic 56708 Version 3.0

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50. A 50-year-old man with amyotrophic lateral sclerosis is admitted to the ICU for the management of respiratory distress. He has been a resident of a skilled nursing facility for 2 years. NKDA. The patient is placed on a mechanical ventilator.

On Hospital Day 5 he develops a fever, leukocytosis, increase in respiratory secretions, pulmonary consolidation on PE, and radiographic infiltrates. The patient is evaluated for antibiotic therapy. Potential infective organisms are MDR bacteria: Gram-negative bacilli, *P. aeruginosa*, pneumococcus, and *S. aureus*. The hospital's antibiogram indicates that MDR *P. aeruginosa* is prevalent in the hospital.

Definition: Antibiogram: a report that shows the susceptibility of pathogenic strains to antibiotics

i) **What are the potential mechanisms of resistance? Highlight all that apply.**

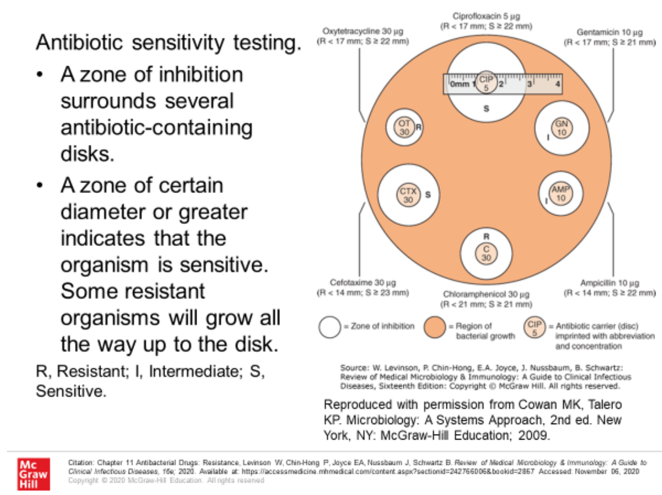
ii) **Which resistance mechanism is most common?**

Deletion of porins
Conditions of low oxygen tension
Expression of MLS_B methylases
Microbial transferases phosphorylate, adenylate, or acetylate the drug
Ribosomal proteins with low affinity for the drug
Thickening of the peptidoglycan wall

Answers (yellow highlights) and Rationale:

Deletion of porins	Aminoglycosides enter the GNB by diffusion through porins in the outer membrane.
Conditions of low oxygen tension	Aminoglycosides require an oxygen-dependent process for transportation across the plasma membrane. This mechanism is not functional in anaerobic bacteria and anaerobic conditions.
Expression of MLS_B methylases	Incorrect: This is a mechanism of macrolides in which the ribosomal protein is methylated, reducing affinity for the drug.
Microbial transferases phosphorylate, adenylate, or acetylate the drug	Plasmid-mediated transferases inactivate the drug by addition of a phosphoryl, adenyl, or acetyl group. This is the main mechanism of resistance.
Ribosomal proteins with low affinity for the drug	Mutations that result in deletion or alteration of the site of action on the 30S ribosomal proteins prevent binding of the drug.
Thickening of the peptidoglycan wall	Incorrect: This is a mechanism of S. aureus resistance to vancomycin. It results from strains that have acquired the D-ala-D-lactate mutation from enterococcus.

Right: An example of a method used for sensitivity/resistance testing:



51. i) What are the three major toxicities associated of aminoglycosides? Short answer

ii) What are the mechanisms of these adverse effects? Short answer

iii) What tests can be used to monitor for these toxicities? Short answer

Rationale:

- Nephrotoxicity: Aminoglycosides accumulate in proximal tubule epithelial cells and cause acute tubular necrosis. The toxicity is often reversible.
  - Closely monitor urine output, serum creatinine, blood urea nitrogen, and AG levels.
- Ototoxicity: Accumulation in the endolymph, which causes vestibular toxicity and dysfunction and toxicity to hair cells and hearing loss. The ototoxicity may be irreversible.
  - Hearing test before, during, and after treatment, particularly in patients receiving AG therapy for >2 weeks; presence of tinnitus and/or vertigo.
- Muscle weakness due to neuromuscular blockade: Rarely, aminoglycosides block presynaptic Ca^{2+} channels, which reduces exocytosis of the neurotransmitter acetylcholine and competitively inhibits the postsynaptic neuromuscular nicotinic receptors at the motor end-plate potential, inhibiting depolarization.
  - Risk factors: AGs may enhance the effects of neuromuscular blocking agents – drug-drug interaction.

    Caution in patients with myasthenia gravis and other neuromuscular disorders – risk of respiratory depression – drug-disease interaction.

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52. Gentamicin- and tobramycin-resistant Gram-negative bacteria are usually susceptible to which alternative drug in the same class? Why?

Short answer

Rationale: Amikacin and to the newest aminoglycoside, plazomicin – recall: plazomicin is indicated for complicated UTIs if no other alternatives are available.

Amikacin and plazomicin structures have changes that confer bacterial susceptibility when bacterial resistance is due to drug structure modifications by phosphorylation, adenylation, and acetylation.

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53. A hospitalized patient is given intravenous linezolid for the treatment of an infection caused by multidrug resistant:

A. *Bacteroides fragilis*
B. *Clostridium difficile*
C. *Escherichia coli*
D. *Streptococcus pneumoniae**
E. *Pseudomonas aeruginosa*

Clue:

The student who recognizes the specific Gram-positive and Gram-negative bacteria will easily come to the correct answer, as only one Gram-positive organism is listed.

Rationale: Linezolid is effective only against Gram-positive aerobes and anaerobes (not *C. difficile*). It is not useful against Gram-negative organisms, *B. fragilis*, or *C. difficile*.

Linezolid activity: MDR Gram-positive bacteria:

streptococci, staphylococci, MRSA, and anaerobic Gram-positive cocci, Gram-positive rods

Bacteriostatic but bactericidal against streptococci

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54. The patient in the previous question is concurrently taking an antidepressant that, together with linezolid, can cause a potentially serious adverse drug event manifested by confusion, tachyarrhythmia, hyperthermia, and seizure. The antidepressant increases levels of 5-hydroxytryptamine in the neuronal synapses in the brain. The antibiotic inhibits a specific enzyme that inactivates this monoamine neurotransmitter, further increasing synaptic levels.

The patient will be monitored for signs and symptoms of:

A. Acute kidney injury
B. Cardiotoxicity
C. Hemolytic anemia
D. Hepatitis
E. Serotonin syndrome*
F. Skeletal muscle toxicity

Hint: Handout p29-30.

Hint: 5-hydroxytryptamine has another name. Look it up.

Rationale: Linezolid is a weak monoamine oxidase (MAO) inhibitor. It can cause increased levels of serotonin in the brain and periphery, which, in turn, can lead to serotonin syndrome.

- Biochemistry: Monoamine oxidase (MAO) is an enzyme that catalyzes deamination (inactivation) of the monoamine neurotransmitters and neuromodulators dopamine, norepinephrine, epinephrine, and serotonin.
- Drug-drug interaction: Antidepressants increase levels of serotonin.

MECHANISM of Serotonin Syndrome Adverse Effect
Inhibiting degradation of serotonin by MAO ↓ results in serotonin accumulation in synapses ↓ excess serotonin causes mild symptoms or potentially fatal hyperthermia and other adverse effects

Leslie Goldstein, Pharm.D.

- **Practice point:** The symptoms are often described as a clinical triad of abnormalities
  - Cognitive effects: headache, agitation hypomania, mental confusion, hallucinations, coma
  - Autonomic effects: trembling, sweating, hyperthermia, vasoconstricition, tachycardia, nausea, diarrhea
  - Somatic effects: myoclonus (muscle twitching), hyperreflexia (manifested by clonus), tremor

What the patient experiences:
S/S are listed for clinical context.
No need to memorize them for this course.

LINEZOLID: WHAT YOU NEED TO KNOW ABOUT LINEZOLID'S ADVERSE EFFECTS
PK: Oral, IV; excellent oral bioavailability; widely distributed, high concentration in CSF; renal and nonrenal clearance; t½ variable, adults ~5 hours **PK-PD profile:** AUC/MIC ratio with moderate post-antibiotic effect for S. aureus but none for S. pneumococcus
MOA: Binds a unique site on the 23S rRNA in the A-site pocket near the 30S ribosome, which prevents protein synthesis initiation by positioning of tRNA and formation of the 70S ribosome **Acquired resistance** remains relatively low. **Intrinsic resistance:** Gram-negative aerobic bacteria (efflux) and anaerobic bacteria (various mechanisms)
Spectrum: Gram-positive aerobic and anaerobic cocci and bacilli, including MDR organisms, and active against *M. tuberculosis* Bactericidal against *Streptococci*. Bacteriostatic against *Staphylococci* and *Enterococci* **Therapeutic uses: Drug-resistant Gram-positive infections**

<table>
<tr><td rowspan="2">Usually well tolerated with short-term use (<14 days)
but can cause nausea, vomiting, diarrhea, headache, rash</td><td colspan="2">Potential serious AEs with long-term use</td></tr>
<tr><td>Therapy duration >14 days
Myelosuppression:
· Thrombocytopenia
· Neutropenia, anemia in patients with underlying myelosuppression</td><td>Therapy duration ≥ 28 days
Mitochondrial toxicities:
· Peripheral neuropathy
(complete recovery may not occur)
· Optic neuritis
(improves with discontinuation of drug)
· Lactic acidosis
(resolves with discontinuation of drug)</td></tr>
<tr><td colspan="3">Drug Interactions
Drug-drug: Weak MAO inhibitor: Serotonin syndrome as above. Occurs usually in 6-8 hours
Drug-disease: Conditions of uncontrolled hypertension
Drug-food: Aged foods containing tyramine</td></tr>
</table>

55. An 83-year-old woman tells her gerontologist about a time many years ago when she had to go to the emergency room because of fatigue, weakness, and mental confusion. They told her she had, “some kind of kidney acidosis”. She was told that it apparently resulted from a drug reaction caused by taking leftover antibiotic “to cure my cold. It must have been in my medicine cabinet for 6 or 7 years. I saved it in case I needed it,” she reported.

Which of the following drugs is associated with this toxicity?

A. Ampicillin
B. Cephalexin
C. Chloramphenicol
D. Erythromycin
E. Tetracycline*

Rationale: Tetracycline

Renal tubular acidosis and Fanconi syndrome (generalized proximal tubule dysfunction) have been attributed to accumulation of tetracycline degradation products in the mitochondria.

This toxicity is unique to expired tetracycline. That is, it is not reported with any of the other drugs in the class or other antibiotics.

Rose BD. Metabolic acidosis. In: Jeffers JD, Novrozov M, editors. Clinical physiology of acid-base and electrolyte disorders. 4th ed. New York: McGraw-Hill; 1994. pp. 540–603. [Google Scholar] [Ref list]

Kim YW. Antimicrobial-induced Electrolyte and Acid-Base Disturbances. Electrolyte Blood Press. 2007 Dec; 5(2): 111–115. PMC3894510; Published online 2007 Dec 31. doi: 10.5049/EBP.2007.5.2.111

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3894510/

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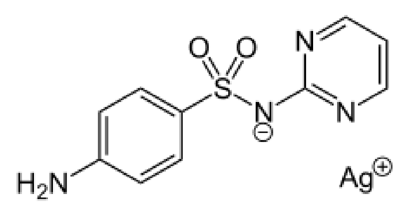
56. A 12-year-old male is admitted for the treatment of second- and third-degree burns on his left arm. His therapy includes an antimicrobial dressing to prevent wound sepsis. The selected sulfonamide is associated with silver, which is released slowly and is selectively toxic to microbes.

What is the sulfonamide component in the dressing?

A. Mafenide
B. Sulfacetamide
C. Sulfadiazine*
D. Sulfadoxine
E. Sulfamethoxazole
F. Trimethoprim



Rationale: C. Sulfadiazine

Silver sulfadiazine topical cream is an adjunctive burn treatment for the prevention and treatment of wound sepsis in patients with 2nd- and 3rd-degree burns.

Ionic silver is slowly released into the wound. The activated silver acts on the bacterial cell wall and cell membrane.

It is bactericidal against many Gram-positive and Gram-negative bacteria and is effective against yeast.

Absorption of silver ions is negligible.

Incorrect options: Students, in addition to silver sulfadiazine, please know the pharmacology of the following sulfonamides.

- Mafenide: Also used topically for the treatment of burns but *it causes intense pain* at sites of application and allergic reactions. Application of the drug over a large burn surface can lead to *appreciable systemic absorption*. The drug and its primary metabolite *inhibit carbonic anhydrase*, which prevents reabsorption of bicarbonate in the proximal tubule. The urine becomes alkaline due to excretion of bicarbonate.
- Sulfacetamide: Ophthalmic drops for the treatment of conjunctivitis and in dermatological products
- Sulfadoxine: Paired with pyrimethamine (and folinic acid = leucovorin) for the prophylaxis and treatment of malaria.
- Sulfamethoxazole: Paired with trimethoprim, which is a dihydrofolate reductase inhibitor not a dihydropteroate synthase inhibitor

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57. MATCHING Differentiating the clinical applications of the aminoglycosides.

Match the drugs with their places in therapy:

Activity against organisms harboring plasmid-mediated transferases	Streptomycin
Combination therapy	Gentamicin
Complicated UTI with no alternative treatment options	Tobramycin
Gram-negative bacilli	Amikacin
Infective endocarditis	Plazomicin
Inhaled for lung infection in cystic fibrosis patients	Neomycin
Intestinal *Entamoeba histolytica* infection	Paromomycin
Not used for systemic infections	All AGs
Pseudomonas aeruginosa	
Reduction of aerobic bowel flora	
Tuberculosis	
Tularemia	

Answers:

Activity against organisms harboring plasmid-mediated transferases	Amikacin and Plazomicin
Combination therapy	ALL

Complicated UTI with no alternative treatment options	Plazomicin
Gram-negative bacilli	ALL except paromomycin, which is an anti-parasitic agent.
Infective endocarditis	Gentamicin with penicillin G or ampicillin for synergistic effect against enteroccocal spp or viridans streptococci Tobramycin is not active.
Inhaled for lung infections in cystic fibrosis patients	Tobramycin against *P. aeruginosa*
Intestinal *Entamoeba histolytica* infection	Paromomycin
Not used for systemic infections	Neomycin Paromomycin’s main place in therapy is topical antimicrobial action within the intestinal lumen for amoebiasis. (Paromomycin may be an option for the treatment of visceral leishmaniasis)
Pseudomonas aeruginosa	Tobramycin has slightly better activity than gentamicin and is usually selected for pseudomonal infections. Gentamicin is also effective.
Reduction of aerobic bowel flora	Neomycin
Tuberculosis	Streptomycin and Amikacin
Tularemia	Streptomycin or gentamicin for severe infection (doxycycline or ciprofloxacin for mild to moderate infection)

- Aminoglycosides are used most widely *in combination* with other antibiotics to treat drug-resistant organism.
- All are effective against susceptible aerobic Gram-negative bacilli.
- For infective endocarditis caused by streptococci or enterococci, gentamicin is preferred – activity against these Gram-positive bacteria when used with a cell wall inhibitor. Gentamicin is generally preferred because of long experience with its use and availability.
- Tobramycin has slightly greater intrinsic activity than gentamicin against *Pseudomonas aeruginosa*. It is often selected, in combination with other agents, for the management of *P. aeruginosa* infections, including the inhaled version for management of pseudomonal lung infection in cystic fibrosis patients.
- Amikacin is reserved for treatment of bacteria resistant to gentamicin and tobramycin.
- Plazomicin is potent activity against Enterobacteriaceae. It has FDA approval for the treatment of complicated UTIs in patients ≥18 years old having limited or no options and caused by *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, and *Enterobacter cloacae*.

- Neomycin is used orally as a bowel cleansing agent prior to elective bowel surgery and topically for dermatologic, ophthalmic, and otic infections.
- Streptomycin is mainly for the treatment of tuberculosis, tularemia, and gentamicin-nonsusceptible endocarditis. Wide resistance in most bacterial species has emerged.
- Paromomycin is used orally as an antiparasitic agent

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58. **Review:** In the treatment of outpatients without comorbidities, what classes of protein synthesis inhibitors would also be effective against susceptible strains of the potential causes of community-acquired pneumonia: "atypicals" and pneumococcus?

Short answer

Answer: Tetracyclines typically doxycycline; Macrolides, typically azithromycin or clarithromycin

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59. **Review:** For outpatients whose community-acquired pneumococcal pneumonia can be treated with an oral agent and who are not allergic, what oral drug in the penicillins class would be useful?

Short answer

Learning tip: SPACED RETRIEVAL PRACTICE
Studying information more than once and leaving considerable time between practice sessions promotes durable learning. The example here is retrieval practice on beta-lactam antibiotics.

Answer: Amoxicillin. High doses if patient is without risk factors for resistant *S. pneumoniae*.

Recall: *S. pneumoniae* resistance mechanism to beta-lactams is expression of PBPs that do not bind the beta-lactams. This resistance can generally be overcome with high-dose amoxicillin or penicillin G (parenteral).

The oral drugs in the penicillins class are:

Natural penicillins subgroup: Penicillin VK

Penicillinase-resistant penicillins subgroup: Dicloxacillin, Cloxacillin (not in US)

Extended-spectrum aminopenicillins subclass: Amoxicillin; Amoxicillin-clavulanate

- Amoxicillin is usually preferred because of its excellent bioavailability and longer half-life and good activity against penicillin-susceptible *S. pneumoniae*.

  Amoxicillin is recommended 3 times daily every 8 hours
- Amoxicillin-clavulanate is recommended for patients with major comorbidities or recent antibiotic use because it has a broader spectrum.
- Penicillin VK is usually recommended for mild to moderated infections caused by Group A *Streptococcus* and Group B *Streptococcus* and certain other uses.

  Amoxicillin is usually preferred. Penicillin VK has relatively poor bioavailability, requires for 4 times daily dosing q6h around the clock, and should be taken with water on an empty stomach.

- The penicillinase-resistant penicillins are generally recommended for methicillin-sensitive *Staphylococcus* infections. Other penicillins are more active against *S. pneumoniae*.

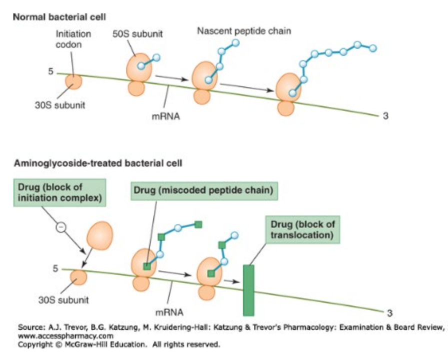
****

SUMMARY OF AMINOGLYCOSIDE PHARMACOLOGY

Structural and pharmacokinetic properties:

- Amino sugars attached by glycosidic linkages
- Polar compounds: Parenteral administration; not absorbed from GI tract
- Limited tissue penetration; do not readily cross the blood-brain barrier
- Excretion via glomerular filtration $\Rightarrow$ excretion is directly proportional to creatinine clearance
- $t½_{elimination}$ 2–3 h; dosage adjustments in patients with renal insufficiency to prevent toxic accumulation
- Monitor plasma levels for safe, effective dosage selection and adjustment
- PK/PD profile: Peak / MIC – concentration dependent killing with prolonger persistent effects (post-antibiotic effect)
- Once-daily dosing (2-3 times daily for traditional dosing regimens)

Mechanism of action:



Katzung & Trevor's Pharmacology: Examination & Board Review, 11e, 2015 Figure 45-1

Entry: Diffusion through porins and oxygen-dependent penetration of bacterial cell membrane

- Block of formation of the initiation complex
- Miscoding of amino acids in the emerging peptide chain due to misreading of the mRNA
- Block of translocation on mRNA
- May also disrupt polysomal structure $\rightarrow$ nonfunctional monosomes

Mechanism of resistance:

- Primary mechanism, Gram-negative bacteria and enterococci:

Plasmid-mediated formation of inactivating enzymes (tranferases) confer resistance by phosphorylation, adenylation, or acetylation

- Amikacin and plazomicin are usually not inactivated by Gram-negative organisms due to modification of its structure. Therefore, it useful for treatment of gentamicin- and tobramycin-resistant infections.
- Streptomycin resistance is common. Resistance mechanisms are typical of the other aminoglycosides. Alteration of the ribosomal binding site develops readily when the drug is used as a single agent.
- Intrinsic resistance: Anaerobic organisms and Gram-positive organisms due to the thick cell wall.
    - Aminoglycoside entry can be enhanced by concomitant cell wall synthesis inhibitors.

Therapeutic uses:

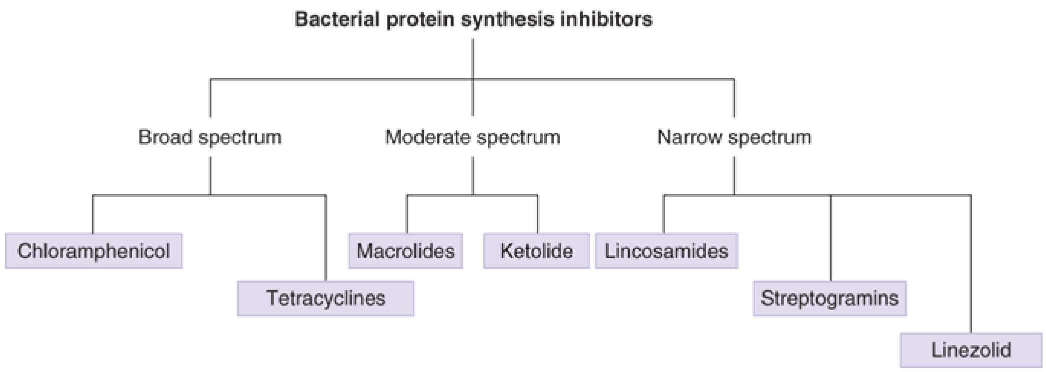
Gentamicin Tobramycin	Treatment of serious infections caused by aerobic Gram-negative bacteria, including *Escherichia coli* and *Enterobacter, Klebsiella, Proteus, Providencia, Pseudomonas*, and *Serratia* species
Amikacin Plazomicin	Treatment of resistant infections.
Synergy may occur when aminoglycosides are used in combination with cell wall synthesis inhibitors. Examples: penicillin G, ampicillin, nafcillin in the treatment of streptococcal and enterococcal infective endocarditis	
Neomycin Paromonycin	Topically Orally: Luminal microbial killing (topical use withing the GI tract)
Streptomycin (wide resistance)	Tuberculosis (*Mycobacterium tuberculosis*) in combination with other antitubercular agents Viridans streptococcal endocarditis, gentamicin-nonsusceptible enterococcal endocarditis Tularemia (*Francisella tularensis*) Plague (*Yersinia pestis*)
Spectinomycin	*Neisseria gonorrhoeae* alternative to ceftriaxone in penicillin-allergic patients (Spectinomycin no longer available in the US.)
Spectinomycin is an amincyclitol related to the aminoglycosides but it lacks aminosugars and glycosidic bonds.	

Adverse effects:

Ototoxicity Auditory and vestibular damage	Ototoxicity and nephrotoxicity are more common: When therapy is continued for more than 5 days At higher doses In the elderly In patients with renal insufficiency
Nephrotoxicity Monitor renal function	
Neuromuscular (NM) blockade	Respiratory paralysis (rare event) with high doses of aminoglycosides Exacerbation of myasthenia gravis and other neuromuscular disease
Streptomycin-related toxicities	In addition to ototoxicity, nephrotoxicity, and NM block, Scotomas (blind spots) Facial paresthesias I.M. injection site reactions

Pregnancy	Relative contraindication due to potential ototoxicity in the fetus/newborn

SUMMARY OF PROTEIN SYNTHESIS INHIBITORS



Source: A.J. Trevor, B.G. Katzung, M. Kruidering-Hall: Katzung & Trevor's Pharmacology: Examination & Board Review, 11th Ed.
www.accesspharmacy.com

- The pharmacokinetic properties vary with the individual classes and drugs within the classes. Most can be administered orally/ Exceptions are tigecycline and streptogramins.
- The drugs all inhibit bacterial protein synthesis by binding to and interfering with ribosomes. They have different sites of action on the ribosome.
- Most are bacteriostatic, but a few are bactericidal against certain organisms.
- Tetracycline and macrolide resistance is common. Tigecycline, eravacycline, and omadacycline structural changes confer activity to certain multidrug resistant Gram-positive and Gram-negative bacteria.
- Adverse effects, including serious and potentially life-threatening toxicities, and drug interactions are drug specific.

Resistance mechanisms:

Tetracyclines:	Tet(K) (Staph resistance to tetracycline Tet(AE) (Gram-negative resistance tetracycline, doxycycline, minocycline chromosomally encoded efflux pumps (Proteus, P. aeruginosa) Tet(M) (Gram-positive resistance by ribosomal protection)
Macrolides	Methylase ribosomal protection: *erm* gene → MLS_B phenotype Efflux pumps: mrsA (S. aureus) mefA (S. pyogenes) mefE (S. pneumoniae)
Clindamycin	Cross-resistance due to methylase: MLS_B phenotype (*erm* gene) Inducible during therapy

Linezolid	Resistance rare to date Decreased affinity for binding site No cross-resistance with other antibiotics
Quinupristin-Dalfopristine	Cross-resistance due to methylase: MLS_B phenotype (*erm* gene) E. faecalis intrinsically resistant (efflux)
Chloramphenicol	Acetyltransferases inactivate the drug

Therapeutic uses:

Tetracyclines Doxycycline and Minocycline	Broad spectrum but much resistance Useful for zoonotic infections, ca-MRSA (ca = community-acquired), Spirochete infections Periodontitis, Acne
Macrolides Azithromycin and Clarithromycin	Respiratory infections, including “atypicals” H. pylori (clarithromycin) sexually transmitted diseases (azithromycin) (Erythromycin is not commonly used.)
Clindamycin	Gram-positive infections and anaerobic infections “above the waist” ca-MRSA but inducible resistance, MLS_B phenotype
Linezolid and other oxazoladinones	Gram-positive MDR infections, i.e. MRSA, VRE, pulmonary tuberculosis Toxicities a concern with long-term use
Quinupristin-Dalfopristin (Streptogramin B and A)	Gram-positive and “atypicals” only Use is reserved for serious MRSA and enterococcus infections
Chloramphenicol	Rickettsial infections in patient who cannot take doxycycline Potential for significant hematologic toxicity

Toxicities

Tetracyclines Doxycycline and Minocycline	GI upsets | deposition in developing bones and teeth | photosensitivity | superinfection Tigecycline: Increased all-cause mortality observed in meta-analyses of phase 3 and 4 clinical trials in tigecycline-treated patients vs. comparator
Macrolides Azithromycin and Clarithromycin	GI upsets | QT prolongation | CYP3A4, P-gp inhibition Azithromycin does not inhibit CYP3A4 Erythromycin: hepatic dysfunction
Clindamycin	GI upsets | *C difficile* colitis
Linezolid and other oxazoladinones	Dose-related anemia | neuropathy, optic neuritis | MAO inhibitor, serotonin syndrome with SSRIs (antidepressants)
Quinupristin-Dalfopristin (Streptogramin B and A)	Infusion-related arthralgia and myalgia | CYP3A4 inhibition | Infusion reactions (phlebitis and pain)

Chloramphenicol	Dose-related anemia, leukopenia, thrombocytopenia │ idiosyncratic aplastic anemia │ gray syndrome

SUMMARY OF FOLATE SYNTHESIS INHIBITORS

Sulfonamides

PK	Modest tissue penetration, hepatic metabolism, and excretion in urine of both intact drug (high concentration) and acetylated metabolites; decreased solubility in acidic urine → drug or metabolites may precipitate (slow acetylators are more subject to adverse effects
Action	Inhibitors of dihydropteroate synthase; compete with the natural substrate PABA
Resistance	Decreased cellular accumulation; enhanced PABA production; alternative folic acid synthesis pathways; modified target; Enterococci, *P. aeruginosa*, anaerobes, spirochetes, and atypical bacteria are intrinsically resistant
Antibacterial effect	Mammalian cells use exogenous folic acid
Spectrum	Broad spectrum but widespread resistance
Therapeutic uses	Limited
Adverse effects	Hypersensitivity reactions, GI side effect, acute hemolysis in patients with G6PD deficiency, crystalluria and hematuria; kernicterus in neonates (displace bilirubin on plasma binding proteins)
Drug interactions	Drugs with narrow therapeutic windows: warfarin, phenytoin, and methotrexate; urinary antiseptic, methenamine → sulfonamide + formaldehyde → insoluble precipitate

Trimethoprim

PK	Good tissue penetration, hepatic metabolism (CYP2C9), high concentrations of intact drug excreted; t½ adults 8-12 hours
Action	Dihydrofolate reductase inhibitor
Resistance	Plasmid-acquired; decreased cellular accumulation; enhanced PABA production; alternative folic acid synthesis pathways; modified target
Antibacterial effect	Bacteriostatic when used alone
Spectrum	Broad spectrum, including caMRSA
Therapeutic uses	Monotherapy is Limited to uncomplicated UTIs Used in combination with sulfamethoxazole
Adverse effects	Hyperkalemia; megaloblastic anemia DDIs: Inhibits CYP2C9
Drug interactions	Potentially significant drug interactions Mechanisms undefined Always check the literature: Drug interaction tools will identify interactions

Trimethoprim-sulfamethoxazole (TMP-SMX, cotrimazole)

- fixed dose combination (1:5 ratio) → Bactericidal effects
- Common therapeutic uses: Urinary tract infections, skin and soft tissue infections, respiratory tract infections, prophylaxis and treatment of *Pneumocysitis jirovecii* infections in immunocompromised patients
- Resistance to TMP-SMX has limited its use against many Enterobacterales

SUMMARY OF FLUORQUINOLONES

Fluoroquinolones Class Properties

PK	Good oral bioavailability (antacids containing multivalent cations may interfere with absorption); penetrate most body tissues. High concentrations in urine (except moxifloxacin – hepatic and biliary elimination)
PK-PD profile	24h AUC/MIC and post-antibiotic effects
Action	Inhibit DNA gyrase (esp Gram-negative organisms) and topoisomerase IV, (esp Gram-positive organisms) → DNA strand breaks / cell death
Resistance	Spontaneous point mutations in the genes encoding the topoisomerases *Resistance can develop during therapy* Reduced intracellular accumulation of the drug; plasmid transfer of genes that protect DNA gyrase
Antibacterial effect	Bactericidal
Spectrum	Broad spectrum, including caMRSA
Therapeutic uses	See the individual agents below
Pregnancy Pediatrics	Avoid due to concerns of cartilage damage in the fetus Breast feeding is not recommended. Avoid in infants and children due to concerns of cartilage damage in the fetus Adverse effects are increased in pediatric patients
Adverse effects	FDA safety alerts: tendinitis, tendon rupture, and worsening of myasthenia gravis, risks of peripheral neuropathy, CNS effects, QTc interval prolongation, dermatologic, and hypersensitivity reactions.

Drug interactions	CYP1A2-mediated (Ciprofloxacin); drugs that prolong QTc interval; glucocorticoids (tendon rupture); divalent cations

The Fluoroquinolones

Ciprofloxacin Ofloxacin	Excellent activity against Gram-negative aerobic bacteria including *P. aeruginosa*; moderate activity against Gram-positive organisms
Levofloxacin	Aerobic Gram-negative bacteria and enhanced Gram-positive activity, including *S. pneumoniae* and other respiratory pathogens
Moxifloxacin	Enhanced Gram-positive activity and reduced activity against Gram-negative bacteria.
Delafloxacin	Acute bacterial skin and skin structure infections (ABSSSI) in adults caused by Gram-positive (including MRSA) and Gram-negative bacteria (including *P. aeruginosa*)

Resources: Links are provided in case you would like to check them out.

Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023; Section VII: Chemotherapy of Infectious Disease, relevant chapters

https://accessmedicine.mhmedical.com.nyit.idm.oclc.org/book.aspx?bookid=3191

Access Medicine Infectious Disease, A Clinical Short Course, 4e, 2020, relevant chapters https://accessmedicine.mhmedical.com/book.aspx?bookid=2816

Access Medicine Jawetz, Adelberg, and Melnick's Medical Microbiology, 28e, 2019; Section III: Bacteriology, relevant chapters https://accessmedicine.mhmedical.com.nyit.idm.oclc, org/Book.aspx?bookid=2629#217768812

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Section VIII: Chemotherapeutic Drugs, relevant chapters

https://accessmedicine.mhmedical.com.nyit.idm.oclc.org/book.aspx?bookid=3382#281742356

Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e 2024; Part VIII: Chemotherapeutic Drugs, relevant Chapters https://accessmedicine.mhmedical.com.nyit.idm.oclc.org/book.aspx?bookid=3461#285552254

Lexicrugs: various drug monographs

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Section VI: Chemotherapeutic Drugs, relevant Chapters https://premiumbasicsciences.lwwhealthlibrary.com.nyit.idm.oclc.org/book.aspx?bookid=3222

ScholarRx Brick: General Microbiology > Antibacterial Drugs > Protein Synthesis Inhibitors https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_MIC0040

The Sanford Guide to Antimicrobial Therapy, 55e, 2025: relevant listings

UpToDate: various articles; literature review current through August 2025